
Safe Position and Attitude Control for Landing on Small Celestial Bodies with Gravitational Uncertainty

Felipe Arenas-Urbe^{1,*}, T. Michael Seigler¹ and Jesse B. Hoagg¹

¹*Department of Mechanical and Aerospace Engineering, University of Kentucky, Lexington, KY, USA*

Correspondence*: felipearur@uky.edu

ABSTRACT

Landing on small celestial bodies is challenging because the spacecraft must satisfy safety and actuator constraints despite gravitational forces that are difficult to model accurately. This article presents an optimal control for safe landing with gravitational uncertainty and strict actuator limits. The approach addresses spacecraft pose control with attitude represented on $SO(3)$, where there are actuator constraints and state constraints on position, attitude, velocity, and angular velocity. The approach combines several key techniques. First, the gravitational force uncertainty is estimated using an extended high-gain observer, and we present a new dynamic upper bound on the estimation error. This gravitational estimate and dynamic bound are then used to compute optimal control forces and torques that satisfy state and actuator constraints while tracking a landing trajectory. Optimal forces and torques are obtained from the closed-form solution to a quadratic program that has a single control barrier function constraint constructed by composing multiple control barrier functions that are designed to enforce each state and actuator constraint. Finally, an optimal allocation maps control forces and torques to actuator commands that satisfy actuator constraints. The method is demonstrated in simulation using 2 vehicle configurations with severe gravitational uncertainty.

Keywords: Spacecraft control, constrained control, optimal control, control barrier functions, polytope projection

1 INTRODUCTION

Small celestial bodies (SCBs) have attracted attention for their scientific value and relevance to planetary defense. Sample return missions provide insights into the formation and composition of celestial bodies. Example missions include NASA OSIRIS-REx, JAXA Hayabusa, and ESA Rosetta. In addition, near-Earth objects motivate active SCB monitoring, and the rapidly developing space economy has generated interest in space mining (Modell, 2025).

Soft landing on SCBs is a proximity maneuver that requires precise control of position and attitude, while satisfying actuator limits and safety constraints (Zhang and Duan, 2023). Spacecraft actuator systems are typically overactuated to provide redundancy as a passive fault-tolerant strategy against actuator failures (Hasan et al., 2022). Nevertheless, soft landing is a challenge because of the uncertain exogenous force due to disturbances such as solar radiation pressure and unmodeled gravitational effects (Mammarella et al., 2025). Error in the mass estimate for a SCB is typically in the range of 10–15% (Bull et al., 2021). In addition, SCBs have irregular shapes and nonuniform mass distribution. To account for these effects, higher-order models can be used (Scheeres, 2012). However, accurately determining model parameters requires high-resolution remote sensing or in-situ measurements that are expensive to acquire (Peñarroya and Paoli, 2019). Even in the best case, the gravitational field model has residual error, which needs to be addressed by the control system.

Spacecraft guidance and control often uses a modular architecture, where the guidance module generates a reference trajectory and the feedback controller is responsible for following the reference. For a review of

trajectory generation for SCB landing, see (Malyuta et al., 2021, Section 3.3). Spacecraft pose control is addressed using a variety of representations, including position-attitude on $\mathbb{R}^3 \times \text{SO}(3)$, dual quaternions, and the special Euclidean group $\text{SE}(3)$ (Zhang and Duan, 2023). Spacecraft pose control methods include integrated, adaptive, and disturbance rejection approaches (Wang and Ji, 2019; Sun and Zheng, 2018; Zhang et al., 2022; Zhang and Duan, 2023).

For position-only control, robust approaches such as H_∞ (Simplício et al., 2018), sliding mode (Fu et al., 2022), and adaptive (Tiwari et al., 2023; Sun and Zheng, 2018) have been developed. Linear extended high-gain observers (EHGO) have also been applied to estimate and compensate for gravitational uncertainties (Dunham et al., 2016). Although these approaches can improve robustness and reference tracking performance, they generally do not provide guarantees of safety constraint satisfaction. On the other hand, model-predictive control (MPC) can explicitly incorporate state and input constraints. MPC methods have been developed for spacecraft proximity operations using classical and dual-quaternion-based formulations (Tiwari et al., 2022; Van Leeuwen et al., 2022; Sun et al., 2021). However, these methods typically rely on approximations and discretizations of the nonlinear spacecraft dynamics. Consequently, guaranteeing recursive feasibility and constraint satisfaction often requires constraint-tightening and robustness mechanisms, which can significantly increase computational complexity.

These limitations motivate the use of control barrier functions (CBFs), which provide a framework to enforce state constraints through forward invariance (Wieland and Allgöwer, 2007; Ames et al., 2019). CBF methods for spacecraft have been developed for orbit transfers (Ong and Ames, 2023), proximity operations (Breedon and Panagou, 2023), and formation flying (Kamat et al., 2026). However, uncertainty in the gravitational field makes SCB landing a challenging application for CBF methods. Although robust CBF approaches (Jankovic, 2018) address model uncertainty, synthesizing a valid CBF subject to actuator constraints is difficult, and model uncertainty only exacerbates this challenge. An alternative approach is to estimate the model uncertainty using a disturbance observer (Das and Murray, 2022; Alan et al., 2023; Sun et al., 2024; Alan et al., 2025; Wang et al., 2025; Daş and Burdick, 2025). However, these approaches rely on bounds on disturbance derivative to characterize observer error, and more importantly, they do not address the challenge of actuator constraints.

This article presents a new control method for safe landing on a SCB with gravitational uncertainty. The approach addresses spacecraft pose control with attitude represented on $\text{SO}(3)$, where there are strict state constraints on position (i.e., glideslope constraint), attitude (i.e., boresight constraint), velocity, and angular velocity, as well as strict actuator constraints. This safe landing approach relies on several key building blocks. First, Section 4 uses a polytope projection to map the set of admissible actuator inputs into a lower-dimensional set of admissible forces and torques. This process transforms the control variable from actuator inputs to control forces and torques, and transforms the actuator constraints into equivalent constraints on the control force and torque. This process is analogous to Durham (1994). This section then presents an optimal control allocation to map admissible forces and torques back to admissible actuator commands. Next, Section 5 presents a nonlinear EHGO to estimate gravitational uncertainty. Notably, we develop a dynamic upper bound on the estimation error, which is critical for guaranteeing safety. This section also uses the gravitational force estimate to develop desired control forces and torques for tracking the reference trajectory, which is assumed to be generated and periodically updated by a guidance module.

Section 6 presents the main contribution of this article, which is an optimal feedback control for safe landing that satisfies all state and actuator constraints. Although the desired control forces and torques achieve landing, they do not generally satisfy state and actuator constraints. Thus, we present a method inspired from (Rabiee and Hoagg, 2024a,b) to construct a composite CBF that can be used to ensure all state and actuator constraints are satisfied. The approach in (Rabiee and Hoagg, 2024a,b) does not address

model uncertainty, so we extend the method to address gravitational uncertainty by using the dynamic upper bound on the estimation error (from Section 5). Note that the dynamic bound is less conservative than a standard robust CBF approach (Daş and Burdick, 2025). Then, the desired forces and torques and the composite CBF are used to construct a closed-form optimal control that achieves landing while satisfying state and actuator constraints. This safe and optimal control is the solution to a quadratic program (QP) that can be expressed in closed form, which is computationally efficient for on-spacecraft implementation.

Finally, Section 7 presents simulation studies using 2 vehicle configurations: a robotic lander similar to the Hayabusa2 mission (Ho et al., 2021), and a crewed landing module similar to the Apollo lunar module (Grumman Aerospace Corporation, 1969). The results demonstrate safe landing under severe gravitational uncertainty where traditional feedback control strategies fail. Some preliminary results related to this article appeared in the conference article (Arenas-Urbe et al., 2025). However, this article goes beyond the preliminary work by addressing position-and-attitude control rather than position-only control.

2 NOTATION

Physical vectors are denoted with bold symbols (e.g., $\mathbf{r}$). A frame is a collection of mutually orthogonal physical unit vectors. If $\mathcal{F}$ is a frame, then $[\mathbf{r}]_{\mathcal{F}}$ is $\mathbf{r}$ resolved in $\mathcal{F}$. The time derivative of $\mathbf{r}$ with respect to the frame $\mathcal{F}$ is denoted by ${}^{\mathcal{F}}\dot{\mathbf{r}}$.

The special orthogonal group $\text{SO}(3)$ is the set of orthogonal matrices in $\mathbb{R}^{3 \times 3}$ with determinant one. The geodesic distance between $P \in \text{SO}(3)$ and $Q \in \text{SO}(3)$ is

$$d(P, Q) \triangleq \arccos \frac{\text{tr } P^{\top} Q - 1}{2} \in [0, \pi],$$

where tr is the trace. Note that $d(P, Q) = 0$ if and only if $P = Q$.

The set of skew-symmetric matrices in $\mathbb{R}^{3 \times 3}$ is denoted by $\text{so}(3)$. Let $\omega \in \mathbb{R}^3$. Then, $[\omega]_{\times} \in \text{so}(3)$ is the skew-symmetric representation of ω , and $(\cdot)^{\vee}$ is the inverse of $[\cdot]_{\times}$, that is, $([\omega]_{\times})^{\vee} = \omega$.

Let $h: \mathbb{R}^n \rightarrow \mathbb{R}^l$ be continuously differentiable. Then, $h': \mathbb{R}^n \rightarrow \mathbb{R}^{l \times n}$ is defined as $h'(x) \triangleq \frac{\partial h(x)}{\partial x}$. The Lie derivatives of h along the vector field of $\psi: \mathbb{R}^n \rightarrow \mathbb{R}^{n \times l}$ is $L_{\psi}h \triangleq h'(x)\psi(x)$. Let $\|\cdot\|$ denote the 2-norm. The boundary of $\mathcal{A} \subseteq \mathbb{R}^n$ is denoted by $\text{bd } \mathcal{A}$.

3 PROBLEM FORMULATION

3.1 Equations of Motion

Consider a SCB (referred to as an asteroid) and a nearby spacecraft (see Figure 1). We assume that the asteroid's mass center is an inertial reference point satisfying Newton's second law. Let $\mathcal{F}_I$ be an inertial frame. Frame $\mathcal{F}_A$ is fixed to the asteroid, and frame $\mathcal{F}_B$ is fixed to the spacecraft. The angular velocity of $\mathcal{F}_A$ relative to $\mathcal{F}_I$ is ω_A . We assume that there are no external moments acting on the asteroid (i.e., ${}^{\mathcal{F}_I}\dot{\omega}_A = \mathbf{0}$), and $\mathcal{F}_I$ is oriented such that $[\omega_A]_{\mathcal{F}_A} = [\omega_A]_{\mathcal{F}_I}$. Let ω_B be the angular velocity of $\mathcal{F}_B$ relative to $\mathcal{F}_I$, and let $\omega_{B/A}$ be the angular velocity of $\mathcal{F}_B$ relative to $\mathcal{F}_A$.

Let $\omega \triangleq [\omega_{B/A}]_{\mathcal{F}_B}$ and $R: [0, \infty) \rightarrow \text{SO}(3)$ be the rotation matrix of $\mathcal{F}_B$ relative to $\mathcal{F}_A$, which satisfies

$$\dot{R}(t) = R(t)[\omega(t)]_{\times}, \quad (1)$$

where $R(0) = R_0$.

Let $\mathbf{J}_B$ be the spacecraft's physical inertia matrix relative to its mass center, and let $\mathbf{T}$ be the resultant torque acting on the spacecraft due to the spacecraft actuators. The spacecraft's angular momentum relative

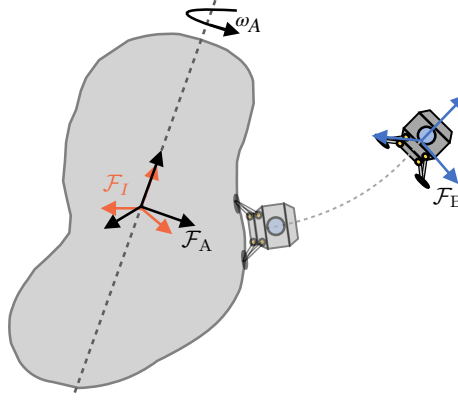


Figure 1. Spacecraft, asteroid, and 3 frames.

to its mass center is

$$\mathbf{H} \triangleq \mathbf{J}_B \boldsymbol{\omega}_B.$$

Since $\boldsymbol{\omega}_B = \boldsymbol{\omega}_A + \boldsymbol{\omega}_{B/A}$ and ${}^{\mathcal{F}_I} \dot{\boldsymbol{\omega}}_A = \mathbf{0}$, the time derivative of $\mathbf{H}$ with respect to $\mathcal{F}_I$ is

$${}^{\mathcal{F}_I} \dot{\mathbf{H}} = \mathbf{J}_B ({}^{\mathcal{F}_B} \dot{\boldsymbol{\omega}}_{B/A} + (\boldsymbol{\omega}_A + \boldsymbol{\omega}_{B/A}) \times \boldsymbol{\omega}_{B/A}) + (\boldsymbol{\omega}_A + \boldsymbol{\omega}_{B/A}) \times \mathbf{J}_B (\boldsymbol{\omega}_A + \boldsymbol{\omega}_{B/A}).$$

Thus, Euler's rotation law implies

$$\mathbf{J}_B ({}^{\mathcal{F}_B} \dot{\boldsymbol{\omega}}_{B/A} + (\boldsymbol{\omega}_A + \boldsymbol{\omega}_{B/A}) \times \boldsymbol{\omega}_{B/A}) + (\boldsymbol{\omega}_A + \boldsymbol{\omega}_{B/A}) \times \mathbf{J}_B (\boldsymbol{\omega}_A + \boldsymbol{\omega}_{B/A}) = \mathbf{T},$$

which is resolved in $\mathcal{F}_B$ as

$$J_B \dot{\omega}(t) + J_B [R(t)^\top \omega_A + \omega(t)]_\times \omega(t) + [R(t)^\top \omega_A + \omega(t)]_\times J_B (R(t)^\top \omega_A + \omega(t)) = T(t), \quad (2)$$

where $\omega_A \triangleq [\omega_A]_{\mathcal{F}_A}$, $J_B \triangleq [\mathbf{J}_B]_{\mathcal{F}_B}$, $T \triangleq [\mathbf{T}]_{\mathcal{F}_B}$, and $\omega(0) = \omega_0$.

The position and velocity of the spacecraft's mass center relative to the asteroid's mass center are $\mathbf{r}$ and $\mathbf{v} \triangleq {}^{\mathcal{F}_I} \dot{\mathbf{r}}$. The spacecraft mass $m > 0$ is assumed constant. Define $r \triangleq [\mathbf{r}]_{\mathcal{F}_A}$, and let $\mathbf{G}(\mathbf{r})$ be the mass-normalized asteroid's gravitational force on the spacecraft. Let $\mathbf{F}$ be the resultant force acting on the spacecraft due to the spacecraft actuators. Thus, Newton's second law implies

$$m {}^{\mathcal{F}_I} \dot{\mathbf{v}} = m \mathbf{G}(\mathbf{r}) + \mathbf{F},$$

which is resolved in $\mathcal{F}_A$ as

$$\dot{r}(t) = v(t), \quad (3)$$

$$\dot{v}(t) = -2[\omega_A]_\times v(t) - [\omega_A]_\times^2 r(t) + G(r) + \frac{1}{m} R(t) F(t), \quad (4)$$

where $v \triangleq [\mathbf{v}]_{\mathcal{F}_A}$, $[\mathbf{G}(\mathbf{r})]_{\mathcal{F}_A} = G(r)$, $F \triangleq [\mathbf{F}]_{\mathcal{F}_B}$, $r(0) = r_0$, and $v(0) = v_0$.

The gravitational force G is uncertain because of uncertainty in mass and mass distribution. Let $G_m : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a known gravitational model, which is 2-times continuously differentiable. Then, the gravitational model error is

$$\Delta(r) \triangleq G(r) - G_m(r).$$

We make the following technical assumption.

ASSUMPTION 1. Δ and Δ' are bounded.

The Newtonian point-mass model is one possible choice for G_m . In cases where more information about the asteroid geometry and composition is available, spherical harmonic models can be a more accurate representation (Scheeres, 2012).

3.2 Asteroid Soft-Landing Problem

The equations of motion (1) to (4) can be written as

$$\dot{x}(t) = f(x(t)) + g(x(t))\mu(t) + Dw(x(t)), \quad (5)$$

where $n = 18$ and

$$x \triangleq \begin{bmatrix} r \\ v \\ \text{vec } R \\ \omega \end{bmatrix}, \quad x_0 \triangleq \begin{bmatrix} r_0 \\ v_0 \\ \text{vec } R_0 \\ \omega_0 \end{bmatrix}, \quad \mu \triangleq \begin{bmatrix} F \\ T \end{bmatrix}, \quad (6)$$

$$f(x) \triangleq \begin{bmatrix} v \\ -2[\omega_A]_{\times} v - [\omega_A]_{\times}^2 r + G_m(r) \\ \text{vec } R[\omega]_{\times} \\ -[R^T \omega_A + \omega]_{\times} \omega - J_B^{-1} ([R^T \omega_A + \omega]_{\times} J_B (R^T \omega_A + \omega)) \end{bmatrix}, \quad (7)$$

$$g(x) \triangleq \begin{bmatrix} 0_{3 \times 3} & 0_{3 \times 3} \\ \frac{1}{m} R & 0_{3 \times 3} \\ 0_{9 \times 3} & 0_{9 \times 3} \\ 0_{3 \times 3} & J_B^{-1} \end{bmatrix}, \quad D \triangleq \begin{bmatrix} 0_{3 \times 3} \\ I_{3 \times 3} \\ 0_{9 \times 3} \\ 0_{3 \times 3} \end{bmatrix}, \quad w(x) \triangleq \Delta(r). \quad (8)$$

The control force F and control torque T depend on $m \geq 6$ actuator inputs (e.g., thrusters) denoted by $u(t) \in \mathcal{U}$, where the set of admissible actuator inputs $\mathcal{U} \subseteq \mathbb{R}^m$ is a convex polytope and

$$\mu = Mu, \quad (9)$$

where $M \in \mathbb{R}^{6 \times m}$ and $\text{rank } M = 6$, which implies that the system is fully actuated or overactuated.

The soft-landing maneuver is a controlled touchdown on the surface of the asteroid, where the spacecraft achieves a final position $r_f \in \mathbb{R}^3$ and final attitude $R_f \in \text{SO}(3)$ with zero translational and angular velocity—all within predefined tolerances. Let $\theta_a \in (0, \pi]$ be the allowable final attitude error, and let $r_a, v_a, \omega_a > 0$ be the allowable landing radius, final speed, and final angular speed, respectively. Then, the set of allowable landing states is

$$L \triangleq \{x \in \mathbb{R}^n : \|r - r_f\| \leq r_a, \|v\| \leq v_a, d(R, R_f) \leq \theta_a, \|\omega\| \leq \omega_a\}.$$

We say that the solution x to (5) to (8) *lands* if there exists $t_1 \in [0, \infty)$ such that $x(t_1) \in L$.

In addition to landing, the spacecraft must satisfy state (e.g., safety) and actuator constraints. Let $\theta_{gs} \in (0, \frac{\pi}{2}]$ be the landing glideslope angle, and consider the glideslope barrier function

$$\psi_1(x) \triangleq \hat{e}_s^T (r - \beta r_f) - \|r - \beta r_f\| \cos \theta_{gs}, \quad (10)$$

where $\hat{e}_s \in \mathbb{R}^3$ is the perpendicular unit vector at the landing position r_f resolved in $\mathcal{F}_A$, and $\beta \triangleq 1 - \frac{r_a}{\|r_f\| \tan \theta_{gs}}$ is a relaxation of the landing position based on the allowable landing radius r_a . The zero-superlevel set of ψ_1 is

$$\Psi_1 \triangleq \{x \in \mathbb{R}^n : \psi_1(x) \geq 0\},$$

which is the set of states that satisfy the glideslope constraint. In this case, the spacecraft is in the allowable approach cone as shown in Figure 2a.

Next, let $\theta_b \in (0, \frac{\pi}{2}]$ be the boresight angle, and consider the boresight barrier function

$$\psi_2(x) \triangleq \hat{e}_b(R)^\top (\beta r_f - r) - \|\beta r_f - r\| \cos \theta_b, \quad (11)$$

where $\hat{e}_b(R) \triangleq -R \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^\top$ is the boresight direction. The zero-superlevel set of ψ_2 is

$$\Psi_2 \triangleq \{x \in \mathbb{R}^n : \psi_2(x) \geq 0\},$$

which is the set of states that satisfy the boresight constraint. In this case, the spacecraft is in the allowable boresight cone, where its attitude is such that it points at the landing position as shown in Figure 2b.

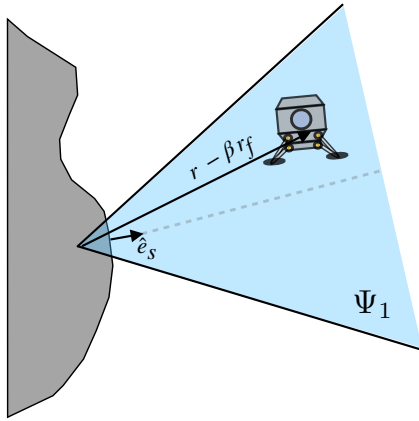


Figure 2a. Zero-superlevel set of ψ_1 .

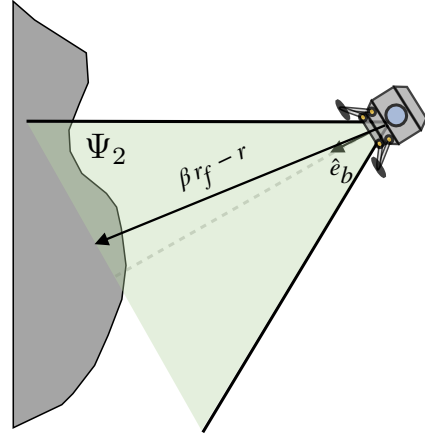


Figure 2b. Zero-superlevel set of ψ_2 .

Figure 2. Glideslope and boresight constraints and their zero-superlevel sets.

We also consider constraints on the translational speed and angular speed. Let $v_m > v_a$ be the maximum allowable translational speed, and let $\omega_m > \omega_a$ be the maximum allowable angular speed. Then, consider the barrier functions

$$\psi_3(x) \triangleq \varphi(\|r - r_f\|^2) v_m^2 + (1 - \varphi(\|r - r_f\|^2)) v_a^2 - \|v\|^2, \quad (12)$$

$$\psi_4(x) \triangleq \varphi(\|r - r_f\|^2) \omega_m^2 + (1 - \varphi(\|r - r_f\|^2)) \omega_a^2 - \|\omega\|^2, \quad (13)$$

where $\varphi : [0, \infty) \rightarrow [0, 1]$ is 2-times continuously differentiable and nondecreasing such that $\varphi(0) = 0$ and $\lim_{a \rightarrow \infty} \varphi(a) = 1$. As the spacecraft approaches the landing position, φ smoothly transitions the speed limits from the maximum allowable values to the allowable values at landing. The following example provides one possible choice for φ .

EXAMPLE 1. Let $k > 0$, and consider

$$\varphi(z) = \begin{cases} kz - \frac{\sin 2\pi kz}{2\pi}, & 0 \leq z < \frac{1}{k}, \\ 1, & z \geq \frac{1}{k}. \end{cases}$$

Figure 3 shows this example of φ for 3 different values of k .

The set of states that satisfy the translational speed and angular speed constraints are given by

$$\Psi_3 \triangleq \{x \in \mathbb{R}^n : \psi_3(x) \geq 0\}, \quad \Psi_4 \triangleq \{x \in \mathbb{R}^n : \psi_4(x) \geq 0\}.$$

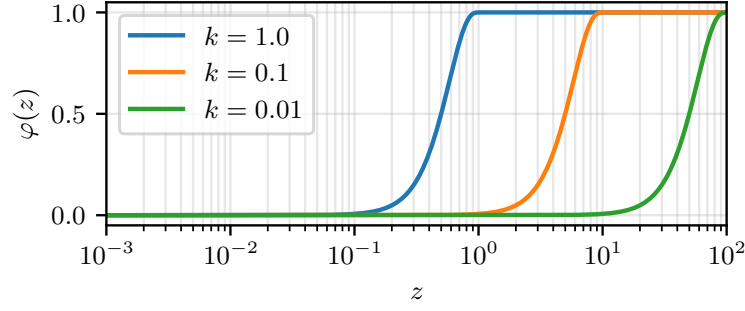


Figure 3. φ given by Example 1 for 3 values of k , which determines the distance at which the speed limits are given by maximum allowable values v_m and ω_m .

Thus, the set of states that satisfy all state constraints is given by

$$\Psi \triangleq \bigcap_{i \in \{1,2,3,4\}} \Psi_i.$$

Then, the control objectives are summarized as:

- (O1) Solution x lands.
- (O2) For all $t \in [0, \infty)$, $x(t) \in \Psi$.
- (O3) For all $t \in [0, \infty)$, $u(t) \in \mathcal{U}$.

Together, these objectives imply the spacecraft lands while satisfying all state and actuator input constraints throughout the trajectory despite the uncertainty in the gravitational force. The approach in this article can be directly extended to include additional state constraints by introducing additional barrier functions (e.g., $\psi_5, \psi_6, \dots$) whose zero-superlevel sets model the allowable states. We assume that for all $x_0 \in \Psi$ and continuous μ , (5) to (8) is forward complete.

To accomplish (O1)–(O3), we consider a reference trajectory for the nominal system (5) to (9) with $w = 0$. Specifically, let $x_r : [0, \infty) \rightarrow \mathbb{R}^n$ and $u_r : [0, \infty) \rightarrow \mathbb{R}^m$ denote the state and control of the reference trajectory. We assume $(x, u) = (x_r, u_r)$ satisfies the nominal differential equation (5) to (9) with $w = 0$ and satisfies (O1)–(O3). In other words, the reference trajectory satisfies the objectives with the nominal gravity model. However, the reference trajectory generally does not satisfy the objective with nonzero gravity uncertainty. This nominal open-loop reference trajectory can be generated using a variety of trajectory optimization methods (e.g., Elango et al. (2025); Sagliano et al. (2024); Reynolds et al. (2020)); however, these trajectory optimization methods are not computationally efficient for recomputing the reference in real time. Thus, this article presents a real-time feedback control for satisfying (O1)–(O3) by tracking the reference trajectory while deviating as needed to address uncertainty and satisfy constraints. We make the technical assumptions that $\mu_r \triangleq M u_r$ is continuously differentiable.

4 PROJECTION OF ACTUATOR INPUTS INTO CONTROL FORCES AND TORQUES

This section transforms the control variable from actuator inputs u to control forces and torques μ . First, we project the set $\mathcal{U}$ of admissible actuator inputs to obtain the set of admissible control forces and torques, and we develop a computable representation of this set. Next, we present a quadratic program to obtain the optimal actuator input from control forces and torques.

4.1 Admissible Control Forces and Torques

The set of admissible control forces and torques is

$$\Phi \triangleq \{Mu : u \in \mathcal{U}\}, \quad (14)$$

and note that $u(t) \in \mathcal{U}$ if and only if $\mu(t) \in \Phi$. Hence, (O3) is equivalent to the condition that for all $t \in [0, \infty)$, $\mu(t) \in \Phi$.

Since $\mathcal{U}$ is a convex polytope, it follows from (Rockafellar, 2015, Thm. 19.3) that Φ is a convex polytope. Thus, there exist $a_1, \dots, a_\ell \in \mathbb{R}^6$ and $b_1, \dots, b_\ell \in \mathbb{R}$, where ℓ is a positive integer, such that

$$\Phi = \{\mu \in \mathbb{R}^6 : \phi_1(\mu) \geq 0, \dots, \phi_\ell(\mu) \geq 0\}, \quad (15)$$

where for $i \in \{1, \dots, \ell\}$,

$$\phi_i(\mu) \triangleq a_i^\top \mu + b_i. \quad (16)$$

In other words, (Rockafellar, 2015, Thm. 19.3) implies that the half-plane constraints on the actuator input can be transformed into half-plane constraints on the control force and torque. Figure 4 illustrates the linear projection of $\mathcal{U}$ through M to obtain Φ . In fact, this linear projection can be performed using the beneath-and-beyond algorithm and double-description algorithm in (Fukuda and Prodon, 1996). First, beneath-and-beyond is used to obtain the set of vertices $\mathcal{V}_u$ of $\mathcal{U}$ from the half-space representation. Then, $\mathcal{V}_\mu \triangleq \{Mu : u \in \mathcal{V}_u\}$ contains the set of vertices of Φ . Finally, double-description is used to obtain the minimal half-space representation (15) and (16) from $\mathcal{V}_\mu$. Note that an upper bound on the number ℓ of half-spaces is given by (Ziegler, 1995, Thm 8.23).

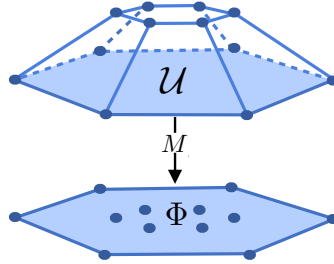


Figure 4. The linear projection of $\mathcal{U}$ through the affine mapping (9) is the set Φ .

4.2 Optimal Actuator Input

To map from admissible control forces and torques to admissible actuator inputs, consider the point-to-set map $\Upsilon : \Phi \rightrightarrows \mathcal{U}$ given by

$$\Upsilon(\mu) \triangleq \{u \in \mathcal{U} : \mu = Mu\}. \quad (17)$$

The following result shows that this set is a convex polytope.

PROPOSITION 1. For all $\mu \in \Phi$, $\Upsilon(\mu)$ is a nonempty convex polytope.

PROOF. Let $\mu \in \Phi$, and it follows from (14) that there exists $u \in \mathcal{U}$ such that $\mu = Mu$. Hence, (17) implies that $u \in \Upsilon(\mu)$, which implies $\Upsilon(\mu)$ is nonempty. Next, since $\Upsilon(\mu)$ is the intersection of the convex polytope $\mathcal{U}$ with the affine subspace defined by the linear equality constraint (9), it follows that $\Upsilon(\mu)$ is a convex polytope. ■

Next, let $Q \in \mathbb{R}^{m \times m}$ be positive definite and $c \in \mathbb{R}^m$. Then, the optimal-actuator-input map is $u_* : \Phi \rightarrow \mathcal{U}$ given by

$$u_*(\mu) \triangleq \underset{\bar{u} \in \Upsilon(\mu)}{\operatorname{argmin}} \bar{u}^\top Q \bar{u} + c^\top \bar{u}. \quad (18)$$

For all $\mu \in \Phi$, the optimizer $u_*(\mu)$ exists and is unique because $\Upsilon(\mu)$ is nonempty and convex and Q is positive definite. Letting $Q = I_m$ and $c = 0$ implies that $u_*(\mu)$ is the minimum-norm actuator input in $\mathcal{U}$ that yields control forces and torques μ .

The quadratic program (18) combined with the computable convex polytope (15) and (16) allows us to address (O3) in the space Φ . Then, (18) is solved to obtain the optimal actuator input. Hence, the remainder of this article treats μ as the control variable.

5 GRAVITY ESTIMATOR, AND DESIRED CONTROL FORCE AND TORQUE

This section presents an EHGO to estimate the gravitational uncertainty and uses this estimate to construct the desired control force and torque that achieve (O1).

5.1 Gravity Estimator

To compensate for gravitational uncertainty, we estimate w using an EHGO method, which is similar to (Dunham et al., 2016). Let $\tau > 0$, and let $\nu : [0, \infty) \rightarrow \mathbb{R}^3$ satisfy

$$\dot{\nu}(t) = -\tau \left[a(x(t)) + \frac{1}{m} R(t) F(t) + \hat{w}(x(t), \nu(t)) \right], \quad (19)$$

where $\nu(0) = \nu_0 \in \mathbb{R}^3$ and

$$a(x) \triangleq -2[\omega_A]_{\times} v - [\omega_A]_{\times}^2 r + G_m(r), \quad (20)$$

$$\hat{w}(x, \nu) \triangleq \tau v + \nu. \quad (21)$$

Note that $\hat{w}$ is the estimate of w , and the error is $\tilde{w}(x, \nu) \triangleq w(x) - \hat{w}(x, \nu)$. Next, let $\delta : [0, \infty) \rightarrow \mathbb{R}$ satisfy

$$\dot{\delta}(t) = -\tau \delta(t) + \frac{c_\delta}{\tau} \|v(t)\|^2, \quad (22)$$

where $\delta(0) = \delta_0$ and $c_\delta > 0$ is such that for all $r \in \mathbb{R}^3$, $c_\delta \geq \|\Delta'(r)\|^2$. Next, define

$$\xi(x, \nu, \delta) \triangleq \delta - \|\tilde{w}(x, \nu)\|^2. \quad (23)$$

The following result shows that δ is an upper bound on the error $\|\tilde{w}\|^2$. The proof is in Appendix A.

PROPOSITION 2. Consider (5) to (8), (19) and (22), where Assumption 1 is satisfied and μ is continuous. Let $\xi(x_0, \nu_0, \delta_0) \geq 0$. Then, for all $t \in [0, \infty)$, $\xi(x(t), \nu(t), \delta(t)) \geq 0$.

Proposition 2 is important because the computable bound δ is used to guarantee state and input constraints despite the gravity uncertainty.

5.2 Desired Control Force

This section presents a desired control force that tracks the reference (x_r, μ_r) and compensates for the uncertainty w . Let x_r and μ_r be written as

$$x_r = \begin{bmatrix} r_r \\ v_r \\ \text{vec } R_r \\ \omega_r \end{bmatrix}, \quad \mu_r = \begin{bmatrix} F_r \\ T_r \end{bmatrix},$$

where $r_r(t) \in \mathbb{R}^3$, $v_r(t) \in \mathbb{R}^3$, $R_r(t) \in \text{SO}(3)$, $\omega_r(t) \in \mathbb{R}^3$, $F_r(t) \in \mathbb{R}^3$ and $T_r(t) \in \mathbb{R}^3$.

Define the position and velocity error

$$e_r(t, x) \triangleq r - r_r(t), \quad e_v(t, x) = v - v_r(t).$$

Then, consider the desired control force

$$F_d(t, x, \nu) \triangleq mR^\top \left(-a(x) - \hat{w}(x, \nu) + a(x_r(t)) + \frac{1}{m}R_r(t)F_r(t) - K_v e_v(t, x) - K_r e_r(t, x) \right), \quad (24)$$

where $K_r, K_v \in \mathbb{R}^{3 \times 3}$ are positive definite. The following result provides convergence properties for e_r and e_v . The proof is in Appendix A.

PROPOSITION 3. Consider (5) to (8), (19) and (22), where Assumption 1 is satisfied, T is continuous, and $F = F_d$. Let $\xi(x_0, \nu_0, \delta_0) \geq 0$, and assume that for all $t \geq 0$, $\|v(t)\| \leq v_m$. Then, there exists $\vartheta > 0$ such that

$$\left\| \begin{bmatrix} e_r(t, x(t)) \\ e_v(t, x(t)) \end{bmatrix} \right\| \rightarrow \left[0, \frac{\vartheta}{\tau} \right] \quad \text{as } t \rightarrow \infty.$$

Proposition 3 shows that the desired control force (24) yields arbitrarily small steady-state error based on design of the EHGO gain τ .

5.3 Desired Control Torque

This section presents a desired control torque that tracks the reference (x_r, μ_r) . Define the rotation and angular velocity errors

$$e_R(t, x) \triangleq \frac{1}{2\sqrt{1 + \text{tr}[R_r(t)^\top R]}} (R_r(t)^\top R - R^\top R_r(t))^\vee, \quad e_\omega(t, x) \triangleq \omega - R^\top R_r(t)\omega_r(t).$$

Then, consider the desired control torque

$$T_d(t, x) \triangleq J_B \left([R^\top R_r(t)\omega_r(t)]_\times J_B R^\top R_r(t)\omega_r(t) + J_B R^\top R_r(t)\dot{\omega}_r(t) - K_R e_R(t, x) - K_\omega e_\omega(t, x) \right), \quad (25)$$

where $K_R, K_\omega \in \mathbb{R}^{3 \times 3}$ are positive definite. The next result is from (Lee, 2012, Proposition 3) and demonstrates that the errors converge to zero.

PROPOSITION 4. Consider (1) and (2), where $T = T_d$. Then, $\lim_{t \rightarrow \infty} e_R(t, x(t)) = 0$ and $\lim_{t \rightarrow \infty} e_\omega(t, x(t)) = 0$.

Let $\mu_d : [0, \infty) \times \mathbb{R}^n \times \mathbb{R}^3 \rightarrow \mathbb{R}^6$ be given by

$$\mu_d(t, x, \nu) \triangleq \begin{bmatrix} F_d(t, x, \nu) \\ T_d(t, x) \end{bmatrix}. \quad (26)$$

Propositions 3 and 4 imply that for a sufficiently large τ , (O1) is satisfied with $\mu = \mu_d$. However, μ_d does not necessarily satisfy (O2). Furthermore, μ_d is not necessarily in Φ , which implies that it cannot necessarily be generated by a control u that satisfies (O3).

6 OPTIMAL CONTROL FOR SAFE LANDING

This section presents a minimum-intervention control to satisfy (O2) and (O3). We first derive higher-order barrier functions to express both objectives as forward invariance of a set. Next, we construct a composite CBF which is then used to compute a control that ensures forward invariance of such set.

6.1 Control Dynamics and Barrier Functions

Since $\phi_1, \dots, \phi_\ell$ depend explicitly on μ , they are not valid candidate CBFs. We adopt the method in (Rabiee and Hoagg, 2024a,b) to transform these input constraints into controller-state constraints by introducing control dynamics. Let the control force and torque μ be the solution to the integrator

$$\dot{\mu}(t) = \zeta(t), \quad (27)$$

where $\mu(0) = \mu_0$ is the initial condition and $\zeta : [0, \infty) \rightarrow \mathbb{R}^6$ is the surrogate control, which is the input to the integrator (27).

The cascade of (5) to (8), (19), (22) and (27) of all system states can be expressed as

$$\dot{\tilde{x}} = \tilde{f}(\tilde{x}) + \tilde{B}\zeta + \tilde{D}w(x) \quad (28)$$

where $\tilde{n} = 28$ and

$$\tilde{x} \triangleq \begin{bmatrix} x \\ \mu \\ \nu \\ \delta \end{bmatrix}, \quad \tilde{x}_0 \triangleq \begin{bmatrix} x_0 \\ \mu_0 \\ \nu_0 \\ \delta_0 \end{bmatrix}, \quad (29)$$

$$\tilde{f}(\tilde{x}) \triangleq \begin{bmatrix} f(x) + g(x)\mu \\ 0_{6 \times 1} \\ -\tau \left[a(x) + \frac{1}{m}RF + \hat{w}(x, \nu) \right] \\ -\tau\delta + \frac{c_\delta}{\tau} \|v\|^2 \end{bmatrix}, \quad \tilde{B} \triangleq \begin{bmatrix} 0_{18 \times 6} \\ I_{6 \times 6} \\ 0_{3 \times 6} \\ 0_{1 \times 6} \end{bmatrix}, \quad \tilde{D} \triangleq \begin{bmatrix} D \\ 0_{6 \times 3} \\ 0_{3 \times 3} \\ 0_{1 \times 3} \end{bmatrix}. \quad (30)$$

Since μ is the state of (27), the input constraints are now state constraints of the cascade (28) to (30). Note that $\phi_1, \dots, \phi_\ell$ have relative degree one with respect to (28) to (30) with input ζ . In contrast, ψ_1 and ψ_2 have relative degree 3 with respect to (28) to (30) with input ζ , and ψ_3 and ψ_4 have relative degree 2. Thus, we adopt a higher-order approach (e.g., Tan et al. (2022)) to construct higher-order candidate CBFs.

First, we address ψ_1 and ψ_2 , which are relative degree 3 with respect to ζ . Let $\beta_1, \beta_2, \gamma_1, \gamma_2 : \mathbb{R} \rightarrow \mathbb{R}$ be extended class- $\mathcal{K}$ functions, where β_1, β_2 are 3-times continuously differentiable and γ_1, γ_2 are 2-times continuously differentiable. Then, for $i \in \{1, 2\}$, consider $\psi_{i,1}, \psi_{i,2} : \mathbb{R}^{\tilde{n}} \rightarrow \mathbb{R}$ defined by

$$\psi_{i,1}(\tilde{x}) \triangleq L_{\tilde{f}}\psi_i(\tilde{x}) + \beta_i(\psi_i(\tilde{x})), \quad (31)$$

$$\psi_{i,2}(\tilde{x}) \triangleq L_{\tilde{f}}\psi_{i,1}(\tilde{x}) + \psi'_{i,1}(\tilde{x})\tilde{D}w(x) + \gamma_i(\psi_{i,1}(\tilde{x})). \quad (32)$$

Next, we address ψ_3 and ψ_4 , which are relative degree 2 with respect to ζ . Let $\beta_3, \beta_4 : \mathbb{R} \rightarrow \mathbb{R}$ be extended class- $\mathcal{K}$ and 2-times continuously differentiable functions. Then, for $i \in \{3, 4\}$, consider $\psi_{i,1} : \mathbb{R}^{\tilde{n}} \rightarrow \mathbb{R}$ defined by

$$\psi_{i,1}(\tilde{x}) \triangleq L_{\tilde{f}}\psi_i(\tilde{x}) + \psi'_i(\tilde{x})\tilde{D}w(x) + \beta_i(\psi_i(\tilde{x})). \quad (33)$$

The intersection of the zero-superlevel sets of the higher-order barrier functions is given by

$$\mathcal{P} \triangleq \{\tilde{x} \in \mathbb{R}^{\tilde{n}} : \psi_{1,2}(\tilde{x}) \geq 0, \psi_{2,2}(\tilde{x}) \geq 0, \psi_{3,1}(\tilde{x}) \geq 0, \psi_{4,1}(\tilde{x}) \geq 0, \phi_1(\mu) \geq 0, \dots, \phi_\ell(\mu) \geq 0\}, \quad (34)$$

and the intersections of all barrier functions is

$$\mathcal{C} \triangleq \{\tilde{x} \in \mathcal{P} : \psi_1(x) \geq 0, \psi_2(x) \geq 0, \psi_3(x) \geq 0, \psi_4(x) \geq 0, \psi_{1,1}(\tilde{x}) \geq 0, \psi_{2,1}(\tilde{x}) \geq 0\}. \quad (35)$$

It follows from (Tan et al., 2022, Prop. 1) that if $\tilde{x}_0 \in \mathcal{C}$ and for all $t \in [0, \infty)$, $\tilde{x}(t) \in \mathcal{P}$, then for all $t \in [0, \infty)$, $\tilde{x}(t) \in \mathcal{C}$, which implies that $x(t) \in \Psi$ and $\mu(t) \in \Phi$. Thus, satisfying (O2) and (O3). However, $\mathcal{P}$ is constructed from the zero-superlevel set of (32) and (33), which are not implementable because they depend on the unknown function w . Thus, we use estimate $\hat{w}$ and bound δ to construct implementable lower bounds. Specifically, for $i \in \{1, 2\}$, consider $\underline{\psi}_{i,2}: \mathbb{R}^{\tilde{n}} \rightarrow \mathbb{R}$ given by

$$\underline{\psi}_{i,2}(\tilde{x}) \triangleq L_{\tilde{f}}\psi_{i,1}(\tilde{x}) + \psi'_{i,1}(\tilde{x})\tilde{D}\hat{w}(x, \nu) + \gamma_i(\psi_{i,1}(\tilde{x})) - \sqrt{\varepsilon + \delta\|\tilde{D}^\top\psi'_{i,1}(\tilde{x})^\top\|^2}, \quad (36)$$

and for $i \in \{3, 4\}$, consider $\underline{\psi}_{i,1}: \mathbb{R}^{\tilde{n}} \rightarrow \mathbb{R}$ given by

$$\underline{\psi}_{i,1}(\tilde{x}) \triangleq L_{\tilde{f}}\psi_i(\tilde{x}) + \psi'_i(\tilde{x})\tilde{D}\hat{w}(x, \nu) + \beta_i(\psi_i(\tilde{x})) - \sqrt{\varepsilon + \delta\|\tilde{D}^\top\psi'_i(\tilde{x})^\top\|^2}, \quad (37)$$

where $\varepsilon > 0$. The next result shows that the computable functions (36) and (37) are lower bounds for (32) and (33).

PROPOSITION 5. Let $\tilde{x} \in \mathbb{R}^{\tilde{n}}$ be such that $\xi(x, \nu, \delta) \geq 0$. Then, for $i \in \{1, 2\}$, $\underline{\psi}_{i,2}(\tilde{x}) \leq \psi_{i,2}(\tilde{x})$, and for $i \in \{3, 4\}$, $\underline{\psi}_{i,1}(\tilde{x}) \leq \psi_{i,1}(\tilde{x})$.

PROOF. Let $i \in \{1, 2\}$, and it follows from (32) and (36) that

$$\begin{aligned} \psi_{i,2}(\tilde{x}) - \underline{\psi}_{i,2}(\tilde{x}) &= \psi'_{i,1}\tilde{D}(w(x) - \hat{w}(x, \nu)) + \sqrt{\varepsilon + \delta\|\tilde{D}^\top\psi'_{i,1}(\tilde{x})^\top\|^2} \\ &\geq \psi'_{i,1}\tilde{D}\tilde{w}(x, \nu) + \sqrt{\delta}\|\tilde{D}^\top\psi'_{i,1}(\tilde{x})^\top\| \\ &\geq -\|\tilde{D}^\top\psi'_{i,1}(\tilde{x})^\top\|\|\tilde{w}(x, \nu)\| + \sqrt{\delta}\|\tilde{D}^\top\psi'_{i,1}(\tilde{x})^\top\| \\ &= \|\tilde{D}^\top\psi'_{i,1}(\tilde{x})^\top\|(\sqrt{\delta} - \|\tilde{w}(x, \nu)\|), \end{aligned}$$

which is nonnegative because $\xi(x, \nu, \delta) \geq 0$. Using the same process for $i \in \{3, 4\}$ we show that $\psi_{i,1}(\tilde{x}) \geq \underline{\psi}_{i,1}(\tilde{x})$, which completes the proof. ■

Since Proposition 5 implies $\xi(x, \nu, \delta) \geq 0$ is sufficient for (36) and (37) to lower bound (32) and (33), define

$$\Xi \triangleq \{\tilde{x} \in \mathbb{R}^{\tilde{n}}: \xi(x, \nu, \delta) \geq 0\}, \quad (38)$$

and note that Proposition 2 implies that Ξ is forward invariant. Next, define

$$\mathcal{P} \triangleq \{\tilde{x} \in \mathbb{R}^{\tilde{n}}: \underline{\psi}_{1,2}(\tilde{x}) \geq 0, \underline{\psi}_{2,2}(\tilde{x}) \geq 0, \underline{\psi}_{3,1}(\tilde{x}) \geq 0, \underline{\psi}_{4,1}(\tilde{x}) \geq 0, \phi_1(\mu) \geq 0, \dots, \phi_\ell(\mu) \geq 0\}, \quad (39)$$

and

$$\underline{\mathcal{C}} \triangleq \mathcal{C} \cap \Xi. \quad (40)$$

The next result shows that forward invariance of $\mathcal{P}$ can ensure forward invariance of $\underline{\mathcal{C}}$.

PROPOSITION 6. Assume Assumption 1 is satisfied. Let $\tilde{x}_0 \in \underline{\mathcal{C}}$ and $t_m \in (0, \infty]$. Assume for all $t \in [0, t_m)$, $\tilde{x}(t) \in \mathcal{P}$. Then, for all $t \in [0, t_m)$, $\tilde{x}(t) \in \underline{\mathcal{C}}$.

PROOF. Let $t_1 \in [0, t_m)$. Since $\tilde{x}_0 \in \underline{\mathcal{C}} = \mathcal{C} \cap \Xi$, Proposition 2 implies that $\tilde{x}(t_1) \in \Xi$. Since, in addition, $\tilde{x}(t_1) \in \mathcal{P}$, it follows from Proposition 5 that $\tilde{x}(t_1) \in \mathcal{P}$. Thus, for all $t \in [0, t_m)$, $\tilde{x}(t) \in \Xi \cap \mathcal{P}$. Since $\tilde{x}_0 \in \mathcal{C}$ and for all $t \in [0, t_m)$, $\tilde{x}(t) \in \mathcal{P}$, it follows from (Tan et al., 2022, Prop. 1) that for all $t \in [0, t_m)$, $\tilde{x}(t) \in \mathcal{C}$, which confirms that $\tilde{x}(t) \in \mathcal{C} \cap \Xi$. ■

Proposition 6 shows that forward invariance of $\underline{\mathcal{P}}$ is sufficient for (O2) and (O3). Motivated by this fact, the next section constructs a candidate CBF whose zero-superlevel set is a subset of $\underline{\mathcal{P}}$.

6.2 Soft-Minimum Relaxed CBF

Let $\rho, k_1, \dots, k_{\ell+4} > 0$, and consider $h : \mathbb{R}^{\tilde{n}} \rightarrow \mathbb{R}$ defined by

$$h(\tilde{x}) \triangleq \text{softmin}_{\rho} \left(k_1 \underline{\psi}_{1,2}(\tilde{x}), k_2 \underline{\psi}_{2,2}(\tilde{x}), k_3 \underline{\psi}_{3,1}(\tilde{x}), k_4 \underline{\psi}_{4,1}(\tilde{x}), k_5 \phi_1(\mu), \dots, k_{\ell+4} \phi_{\ell}(\mu) \right), \quad (41)$$

where

$$\text{softmin}_{\rho}(z_1, \dots, z_N) \triangleq -\frac{1}{\rho} \log \sum_{i=1}^N e^{-\rho z_i} \quad (42)$$

is the log-sum-exponential *soft minimum*. The zero-superlevel set of h is

$$\mathcal{H} \triangleq \{\tilde{x} \in \mathbb{R}^{\tilde{n}} : h(\tilde{x}) \geq 0\}. \quad (43)$$

It follows from (Safari and Hoagg, 2025, Prop. 2) that $\mathcal{H} \subset \underline{\mathcal{P}}$. In fact, as $\rho \rightarrow \infty$, $\mathcal{H} \rightarrow \underline{\mathcal{P}}$. The soft minimum (42) is a continuously differentiable lower bound on the minimum. The worst case conservativeness of the soft minimum in comparison to the minimum is $(\log N)/\rho$ (Rabiee and Hoagg, 2025). Hence, ρ can be selected sufficiently large to limit conservativeness based on N and the order of magnitude of the arguments of the soft minimum. However, selecting ρ is a trade off between conservativeness of the soft-minimum approximation and the magnitude of h' at points where the minimum is not differentiable (Rabiee and Hoagg, 2025). The positive constants $k_1, \dots, k_{\ell+4} > 0$ are selected to normalize the magnitude of the arguments of (41).

Let $\alpha : \mathbb{R} \rightarrow \mathbb{R}$ be locally Lipschitz and nondecreasing such that $\alpha(0) = 0$. Then, consider the constraint function $b : \mathbb{R}^{\tilde{n}} \times \mathbb{R}^6 \times \mathbb{R} \rightarrow \mathbb{R}$ given by

$$b(\tilde{x}, \bar{\zeta}, \bar{\kappa}) \triangleq h'(\tilde{x}) \left[\tilde{f}(\tilde{x}) + \tilde{B}\bar{\zeta} + \tilde{D}w(x) \right] + \alpha(h(\tilde{x})) + \bar{\kappa}h(\tilde{x}), \quad (44)$$

where $\bar{\zeta}$ is the surrogate control variable and $\bar{\kappa}$ is the slack variable. Note that $b(\tilde{x}, \bar{\zeta}, \bar{\kappa}) \geq 0$ is a CBF constraint that can be used to select a surrogate control that makes $\mathcal{H}$ forward invariant. However, b depends on the unknown function w , which implies that $b(\tilde{x}, \bar{\zeta}, \bar{\kappa}) \geq 0$ cannot be implemented. Thus, we use estimate $\hat{w}$ and bound δ to construct the implementable lower bound $\underline{b} : \mathbb{R}^{\tilde{n}} \times \mathbb{R}^6 \times \mathbb{R} \rightarrow \mathbb{R}$ defined as

$$\underline{b}(\tilde{x}, \bar{\zeta}, \bar{\kappa}) \triangleq h'(\tilde{x}) \left[\tilde{f}(\tilde{x}) + \tilde{B}\bar{\zeta} + \tilde{D}\hat{w}(x, \nu) \right] - \sqrt{\delta} \|\tilde{D}^{\top} h'(\tilde{x})^{\top}\| + \alpha(h(\tilde{x})) + \bar{\kappa}h(\tilde{x}). \quad (45)$$

The next result shows that $\underline{b}$ is a lower bound on b . The proof is similar to that of Proposition 5.

PROPOSITION 7. Let $(\tilde{x}, \bar{\zeta}, \bar{\kappa}) \in \Xi \times \mathbb{R}^6 \times \mathbb{R}$. Then, $\underline{b}(\tilde{x}, \bar{\zeta}, \bar{\kappa}) \leq b(\tilde{x}, \bar{\zeta}, \bar{\kappa})$.

Since $b(\tilde{x}, \bar{\zeta}, \bar{\kappa}) \geq 0$ can be used to select a surrogate control that makes $\mathcal{H} \subset \underline{\mathcal{P}}$ forward invariant, it follows from Propositions 6 and 7 that $\underline{b}(\tilde{x}, \bar{\zeta}, \bar{\kappa}) \geq 0$ can be used to select a surrogate control that makes $\underline{\mathcal{C}}$ forward invariant, thus satisfying (O2) and (O3). The next result provides a sufficient condition such that $\underline{b}(\tilde{x}, \bar{\zeta}, \bar{\kappa}) \geq 0$ is feasible for all $\tilde{x} \in \mathbb{R}^{\tilde{n}}$.

PROPOSITION 8. Assume that for all $\tilde{x} \in \text{bd } \mathcal{H}$, $\frac{\partial h}{\partial \mu}(\tilde{x}) \neq 0$. Then, for all $\tilde{x} \in \mathbb{R}^{\tilde{n}}$, there exist $(\bar{\zeta}, \bar{\kappa}) \in \mathbb{R}^6 \times \mathbb{R}$ such that $\underline{b}(\tilde{x}, \bar{\zeta}, \bar{\kappa}) \geq 0$.

PROOF. We consider 2 cases: $\tilde{x} \in \text{bd } \mathcal{H}$, and $\tilde{x} \in \mathbb{R}^{\tilde{n}} \setminus \text{bd } \mathcal{H}$. First, let $\tilde{x} \in \text{bd } \mathcal{H}$. Let $\bar{\kappa} \in \mathbb{R}$ and

$$\bar{\zeta} = \frac{-h'(\tilde{x}) \left[\tilde{f}(\tilde{x}) + \tilde{D}\hat{w}(x, \nu) \right] + \sqrt{\delta} \|\tilde{D}^\top h'(\tilde{x})^\top\| \left[\frac{\partial h}{\partial \mu}(\tilde{x}) \right]^\top}{\left\| \frac{\partial h}{\partial \mu}(\tilde{x}) \right\|^2}$$

Then, (45) yields $\underline{b}(\tilde{x}, \bar{\zeta}, \bar{\kappa}) = 0$.

Next, let $\tilde{x} \in \mathbb{R}^{\tilde{n}} \setminus \text{bd } \mathcal{H}$. Let $\bar{\zeta} \in \mathbb{R}^6$ and

$$\bar{\kappa} = \frac{-h'(\tilde{x}) \left[\tilde{f}(\tilde{x}) + \tilde{B}\bar{\zeta} + \tilde{D}\hat{w}(x, \nu) \right] + \sqrt{\delta} \|\tilde{D}^\top h'(\tilde{x})^\top\| - \alpha(h(\tilde{x}))}{h(\tilde{x})}.$$

Then, (45) yields $\underline{b}(\tilde{x}, \bar{\zeta}, \bar{\kappa}) = 0$. ■

Based on Proposition 8, we make the following assumption.

ASSUMPTION 2. For all $\tilde{x} \in \text{bd } \mathcal{H}$, $\frac{\partial h}{\partial \mu}(\tilde{x}) \neq 0$.

6.3 Closed-Form Optimal Control

This section presents a real-time feedback control that satisfies (O1)–(O3). First, consider the desired surrogate control $\zeta_d: [0, \infty) \times \mathbb{R}^{\tilde{n}} \rightarrow \mathbb{R}^6$ defined by

$$\zeta_d(t, \tilde{x}) \triangleq \dot{\mu}_d(t, x, \nu) - \sigma(\mu - \mu_d(t, x, \nu)), \quad (46)$$

where $\sigma > 0$. If $\zeta = \zeta_d$, then the solution μ to the integrator (27) converges exponentially to μ_d (Rabiee and Hoagg, 2024a, Prop. 8). Since the integral of the desired surrogate control ζ_d converges exponentially to the desired control force and torque μ_d , it follows from Propositions 3 and 4 that for a sufficiently large τ , (O1) is satisfied for $\zeta = \zeta_d$. However, μ_d does not necessarily satisfy (O2), and μ_d cannot necessarily be generated by a control u that satisfies (O3). Thus, we consider the minimum-intervention cost function

$$J(t, \tilde{x}, \bar{\zeta}, \bar{\kappa}) \triangleq \|\bar{\zeta} - \zeta_d(t, \tilde{x})\|^2 + \gamma \bar{\kappa}^2, \quad (47)$$

where $\gamma > 0$. The objective is to find $(\bar{\zeta}, \bar{\kappa})$ That minimize J subject to the constraint $\underline{b}(\tilde{x}, \bar{\zeta}, \bar{\kappa}) \geq 0$.

For each $\tilde{x} \in \mathbb{R}^{\tilde{n}}$, the minimizer of (47) subject to $\underline{b}(\tilde{x}, \bar{\zeta}, \bar{\kappa}) \geq 0$ can be derived from the first-order necessary conditions for optimality (e.g., Rabiee and Hoagg (2024a); Safari and Hoagg (2025); Wieland and Allgöwer (2007)). Specifically, the optimal surrogate control $\zeta_*: [0, \infty) \times \mathbb{R}^{\tilde{n}} \rightarrow \mathbb{R}^6$ is given by

$$\zeta_*(t, \tilde{x}) \triangleq \zeta_d(t, \tilde{x}) + \lambda(t, \tilde{x}) \frac{\partial h}{\partial \mu}(\tilde{x})^\top, \quad (48)$$

where $\lambda: [0, \infty) \times \mathbb{R}^{\tilde{n}} \rightarrow \mathbb{R}$ is defined by

$$\lambda(t, \tilde{x}) \triangleq \max \left\{ 0, \frac{-\underline{b}(\tilde{x}, \zeta_d(t, \tilde{x}), 0)}{\left\| \frac{\partial h}{\partial \mu}(\tilde{x})^\top \right\|^2 + \gamma^{-1} h(\tilde{x})^2} \right\}. \quad (49)$$

In addition, the optimal slack variable $\kappa_*: [0, \infty) \times \mathbb{R}^{\tilde{n}} \rightarrow \mathbb{R}$ is given by

$$\kappa_*(t, \tilde{x}) \triangleq \gamma^{-1} h(\tilde{x}) \lambda(t, \tilde{x}). \quad (50)$$

The parameter γ weights κ in the cost (47) and influences control aggressiveness. The next result shows that the control ζ_* and κ_* satisfy the safety constraint $\underline{b}(x, \zeta_*, \kappa_*) \geq 0$, and provides useful properties regarding the continuity of ζ^* , κ^* and λ .

PROPOSITION 9. If Assumption 2 is satisfied, then, the following statements hold:

- (a) For all $(t, x) \in [0, \infty) \times \mathbb{R}^{\tilde{n}}$, $\underline{b}(x, \zeta_*(t, \tilde{x}), \kappa_*(t, \tilde{x})) \geq 0$.
- (b) ζ_* , κ_* and λ are continuous on $[0, \infty) \times \mathbb{R}^{\tilde{n}}$.
- (c) Assume that h' is locally Lipschitz on $\mathbb{R}^{\tilde{n}}$. Then, ζ_* , κ_* and λ are locally Lipschitz on $[0, \infty) \times \mathbb{R}^{\tilde{n}}$.

PROOF. To prove (a), substituting (48) and (50) into (45) yields

$$\begin{aligned} \underline{b}(\tilde{x}, \zeta_*(t, \tilde{x}), \kappa_*(t, \tilde{x})) &= h'(\tilde{x}) \left[\tilde{f}(\tilde{x}) + \tilde{B} \left(\zeta_d(t, \tilde{x}) + \lambda(t, \tilde{x}) \frac{\partial h}{\partial \mu}(\tilde{x})^\top \right) + \tilde{D} \hat{w}(x, \nu) \right] \\ &\quad - \sqrt{\delta} \|\tilde{D}^\top h'(\tilde{x})^\top\| + \alpha(h(\tilde{x})) + \gamma^{-1} \lambda(t, \tilde{x}) h(\tilde{x})^2 \\ &= \underline{b}(\tilde{x}, \zeta_d(t, \tilde{x}), 0) + \lambda(t, x) \left(\left\| \frac{\partial h}{\partial \mu}(\tilde{x})^\top \right\|^2 + \gamma^{-1} h(\tilde{x})^2 \right) \\ &= \max\{0, \underline{b}(\tilde{x}, \zeta_d(t, \tilde{x}), 0)\} \\ &\geq 0. \end{aligned}$$

To prove (b), since (16), (36) and (37) are continuously differentiable on $\mathbb{R}^{\tilde{n}}$, it follows from (41) and (42) that h is continuously differentiable on $\mathbb{R}^{\tilde{n}}$. Thus, h' is continuous on $\mathbb{R}^{\tilde{n}}$. Furthermore, since μ_r is continuously differentiable on $[0, \infty)$, it follows from (26), (27) and (46) that ζ_d is continuous on $[0, \infty) \times \mathbb{R}^{\tilde{n}}$. Since, in addition, the right hand side of (19) and (22) are locally Lipschitz on $\mathbb{R}^{\tilde{n}}$, it follows that $\hat{w}$ and δ are continuous on $[0, \infty)$. Hence, it follows from (45) that $\underline{b}(\tilde{x}, \zeta_d(t, \tilde{x}), 0)$ is continuous on $[0, \infty) \times \mathbb{R}^{\tilde{n}}$. Moreover, it follows from (49) and Assumption 2 that λ is continuous on $[0, \infty) \times \mathbb{R}^{\tilde{n}}$, which combined with (48) and (50) implies that ζ_* and κ_* are continuous on $[0, \infty) \times \mathbb{R}^{\tilde{n}}$.

To prove (c), note that $\tilde{f}$ is locally Lipschitz on $\mathbb{R}^{\tilde{n}}$ and ζ_d is locally Lipschitz in $\tilde{x}$ on $\mathbb{R}^{\tilde{n}}$. Since, in addition, h and h' are locally Lipschitz on $\mathbb{R}^{\tilde{n}}$, it follows from (45) and (49) that λ is locally Lipschitz on $\mathbb{R}^{\tilde{n}}$, which combined with (48) and (50) implies that ζ_* and κ_* are locally Lipschitz on $\mathbb{R}^{\tilde{n}}$. ■

The following theorem is the main result on constraint satisfaction.

THEOREM 1. Consider (28) to (30), where Assumptions 1 and 2 are satisfied. Let $\zeta = \zeta_*$, where ζ_* is given by (48) and (49). Assume that h' is locally Lipschitz on $\mathbb{R}^{\tilde{n}}$. Then, for all $\tilde{x}_0 \in \mathcal{C}$, the following statements hold:

- (a) There exists a maximum $t_m \in (0, \infty]$ such that (28) to (30) with $\zeta = \zeta_*$ has a unique solution on $[0, t_m)$.
- (b) For all $t \in [0, t_m)$, $\tilde{x}(t) \in \mathcal{C}$.
- (c) For all $t \in [0, t_m)$, $x(t) \in \Psi$ and $\mu(t) \in \Phi$.
- (d) If $t_m = \infty$, then v , R , ω , ν and δ are bounded.

PROOF. To prove (a), it follows from (b) and (c) of Proposition 9 that ζ_* is continuous in t and locally Lipschitz in $\tilde{x}$. Since, in addition, $\tilde{f}$ is locally Lipschitz on $\mathbb{R}^{\tilde{n}}$, it follows from (Khalil, 2002, Theorem 3.1) that (28) to (30) with $\zeta = \zeta_*$ has a unique solution on $[0, t_m)$.

To prove (b), let $\tilde{x}_0 \in \mathcal{C}$, and let $\eta: [0, t_m) \rightarrow \mathbb{R}$ satisfy $\dot{\eta}(t) = -\alpha(\eta(t)) - \beta(t)\eta(t)^2$, where $\eta(0) = h(\tilde{x}_0)$ and $\beta(t) \triangleq \gamma^{-1} \lambda(t, \tilde{x}(t))$. Since α is nondecreasing, $\alpha(0) = 0$, and $\eta(0) = h(\tilde{x}_0) \geq 0$, it follows that

for all $t \in [0, t_m)$, $\eta(t) \geq 0$. Moreover, it follows from (28), (44), (45), (48) and (50), Proposition 7, Proposition 2 and (a) of Proposition 9 that for all $t \in [0, t_m)$,

$$\begin{aligned} \frac{d}{dt}h(\tilde{x}(t)) &= h'(\tilde{x}(t)) \left[\tilde{f}(\tilde{x}(t)) + \tilde{B}\tilde{\zeta}_*(t, \tilde{x}(t)) + \tilde{D}w(x(t)) \right] \\ &= b(\tilde{x}(t), \zeta_*(t, \tilde{x}(t)), \kappa_*(t, \tilde{x}(t))) - \alpha(h(\tilde{x}(t))) - \kappa_*(t, \tilde{x}(t))h(\tilde{x}(t)) \\ &\geq \underline{b}(\tilde{x}(t), \zeta_*(t, \tilde{x}(t)), \kappa_*(t, \tilde{x}(t))) - \alpha(h(\tilde{x}(t))) - \kappa_*(t, \tilde{x}(t))h(\tilde{x}(t)) \\ &\geq -\alpha(h(\tilde{x}(t))) - \kappa_*(t, \tilde{x}(t))h(\tilde{x}(t)) \\ &= -\alpha(h(\tilde{x}(t))) - \beta(t)h(\tilde{x}(t))^2. \end{aligned}$$

Hence, the comparison lemma (Khalil, 2002, Lemma 3.4) implies that for all $t \in [0, t_m)$, $h(\tilde{x}(t)) \geq \eta(t) \geq 0$. Thus, for all $t \in [0, t_m)$, $\tilde{x}(t) \in \mathcal{H} \subset \mathcal{P}$, and Proposition 6 implies that for all $t \in [0, t_m)$, $\tilde{x}(t) \in \mathcal{C}$.

To prove (c), since for all $t \in [0, t_m)$, $\tilde{x}(t) \in \mathcal{C}$, it follows from (35), (39) and (40) that for all $t \in [0, t_m)$, $x(t) \in \Psi$ and $\mu(t) \in \Phi$.

To prove (d), since for all $t \in [0, t_m)$, $x(t) \in \Psi$, it follows that for all $t \in [0, t_m)$, $x(t) \in \Psi_3$ and $x(t) \in \Psi_4$. Then, it follows from (12) and (13) that $v(t)$ and $\omega(t)$ are bounded. Moreover, since $R(t) \in \text{SO}(3)$, it follows that $R(t)$ is bounded. Furthermore, it follows from Proposition 2 that $\nu(t)$ and $\delta(t)$ are bounded. ■

Theorem 1 shows that for all $t \in [0, t_m)$, $x(t) \in \Psi$ and $\mu(t) \in \Phi$, thus, achieving (O2) while producing a control μ that approximates μ_d , which is designed to achieve (O1). Moreover, the optimal allocation (18) can be used to compute an actuator input such that for all $t \in [0, t_m)$, $u_*(\mu(t)) \in \mathcal{U}$, thus, achieving (O3). The complete feedback control architecture is shown in Figure 5.

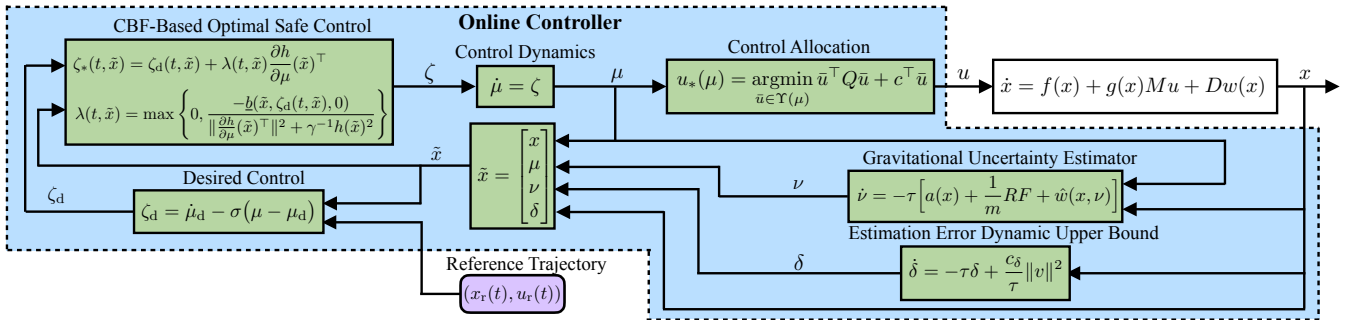


Figure 5. Optimal and safe control with state and input constraints under gravitational uncertainty. Control minimizes cost subject to CBF constraint using the composite CBF h .

7 SIMULATIONS

We consider 2 spacecraft configurations: a robotic lander Ho et al. (2021) and a crewed landing module Grumman Aerospace Corporation (1969). The landing tolerances are $r_a = 2.0$ m, $v_a = 0.5$ m/s, $\theta_a = 0.2$ rad, and $\omega_a = 0.05$ rad/s. For the constraints, we let $\theta_{gs} = \pi/3$, $\theta_b = \pi/3$, $v_m = 12$ m/s, and $\omega_m = 0.1$ rad/s, and φ is given by Example 1 with $k = 10^{-5}$.

The unknown gravitation force G is a spherical harmonics model, whereas the known model G_m is a Newtonian point-mass model with 10% less mass than the true value. The spherical harmonics model is implemented using (Arenas-Urbe, 2026). Reference trajectories (x_r, u_r) are obtained with the trajectory generation algorithm (Malyuta et al., 2021).

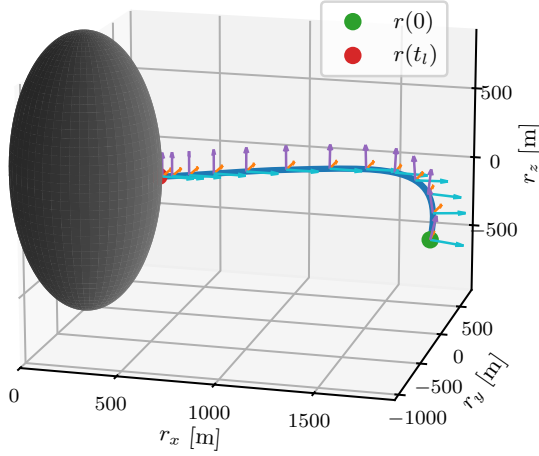


Figure 6a. Spacecraft trajectory and body-frame attitude.

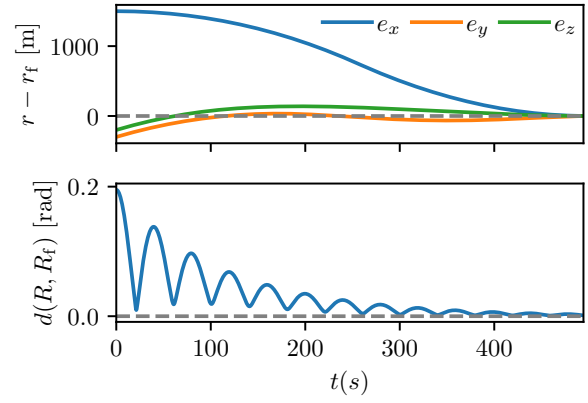


Figure 6b. Distance $r - r_f$ to landing, and geodesic distance $d(R, R_f)$ to landing attitude. Note e_x, e_y, e_z are the entries of $r - r_f$.

Figure 6. Robotic lander soft-landing maneuver on elliptical asteroid.

The EHGO (19) and dynamic upper bound (22) are implemented with $\tau = 10$, $c_\delta = 10^{-3}$, $\nu_0 = -\tau\nu_0$, and $\delta_0 = 10^{-10}$. We implement the control (18), (24) to (27), (46), (48) and (49) with $\sigma = 5$, $\mu_0 = \mu_r(0)$, $\rho = 40$, $\alpha(h) = h$, $\gamma = 10^{20}$, $\beta_1(\psi_1) = 0.1\psi_1$, $\gamma_1(\psi_{1,1}) = 0.1\psi_{1,1}$, $\beta_2(\psi_2) = \psi_2$, $\gamma_2(\psi_{2,1}) = \psi_{2,1}$, $\beta_3(\psi_3) = \psi_3$ and $\beta_4(\psi_4) = 10\psi_4$. The control is implemented with an update rate of 25 Hz. The project repository is at ¹.

EXAMPLE 2. We consider a robotic lander similar to the Hayabusa2 mission (Ho et al., 2021) with mass of $m = 600$ kg and inertia

$$J = \begin{bmatrix} 180 & -5 & -2 \\ -5 & 180 & -8 \\ -2 & -8 & 150 \end{bmatrix} \text{ kg} \cdot \text{m}^2.$$

The spacecraft has 4 reaction wheels in a pyramidal arrangement, and 3 bi-directional thrusters in an orthogonal configuration, which yields the allocation matrix

$$M = \frac{1}{3} \begin{bmatrix} 0 & 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3 \\ 3 & -1 & -1 & -1 & 0 & 0 & 0 \\ -1 & 3 & -1 & -1 & 0 & 0 & 0 \\ -1 & -1 & 3 & -1 & 0 & 0 & 0 \end{bmatrix}.$$

Each reaction wheel has a torque limit of ± 0.2 N-m, and each thruster has a force limit of ± 20 N. Thus, the admissible actuator input set $\mathcal{U}$ is a hyper-cube with 14 half-spaces, and the resulting admissible control-force-and-torque set Φ has 18 half-spaces. Control gains are $K_r = 0.1 I_3$, $K_v = 0.632 I_3$, $K_R = \text{diag}(400, 400, 335)$ and $K_\omega = \text{diag}(536, 536, 448)$.

The asteroid is ellipsoidal with semi-axes lengths of 1000 m, 400 m, and 400 m, and uniform mass distribution with density 1380 kg/m^3 , which is typical for a C-type asteroid. The asteroid rotates about its principal axis associated with the largest moment of inertia with synodic period 4 h.

The initial conditions are $r_0 = [1900 \ -300 \ -200]^\top \text{ m}$, $v_0 = [0 \ 4 \ 4]^\top \text{ m/s}$, $\omega_0 = 0 \text{ rad/s}$, and R_0 is the rotation corresponding to a 1-2-3 Euler sequence with angles $(0.174, 0.087, 0) \text{ rad}$. The final position is $r_f = [400 \ 0 \ 0]^\top \text{ m}$, and the final attitude R_f is the rotation corresponding to a 1-2-3 Euler sequence with angles $(0, \pi/2, 0) \text{ rad}$.

¹ https://github.com/FelipeArenasUribe/Asteroid_Safe_Landing

Landing is achieved at $t_1 = 494.4$ s, where $\|r - r_f\| = 0.786$ m, $\|v - v_f\| = 0.147$ m/s, $d(R, R_f) = 0.001$ rad, $\|\omega - \omega_f\| = 0.001$ rad/s. Figure 6a shows the spacecraft trajectory, and Figure 6b shows the distance $r - r_f$ and geodesic $d(R, R_f)$ throughout the maneuver. Figure 7a shows that the composite CBF h is nonnegative throughout the maneuver, which implies that $\psi_1, \psi_2, \psi_3, \psi_4, \phi_1, \dots, \phi_{18}$ are nonnegative. Thus, $x(t) \in \Psi$ and $u(t) \in \mathcal{U}$ for $t \in [0, t_1]$. Figure 7a also shows that λ is positive for $t \in [467, 493]$, which indicates that the desired control forces and torques are modified during this time interval to satisfy state and actuator constraints. Figures 7b and 7c confirm that the state and actuator constraints are satisfied throughout the maneuver. Figure 7d shows that the magnitude of the estimation error $\tilde{w}$ and satisfies the upper bound $\sqrt{\delta}$. The magnitude of $\tilde{w}$ increases as the spacecraft approaches the asteroid because the influence of higher-order gravitational effects increases.

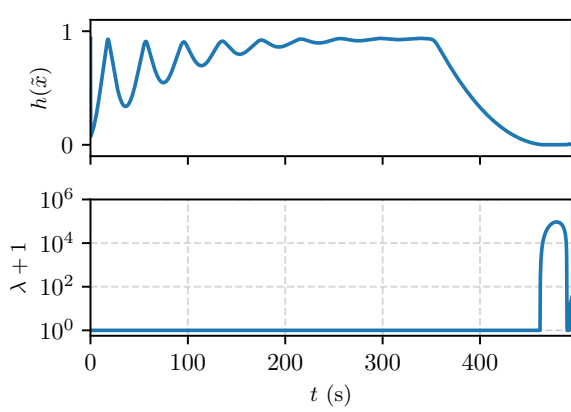


Figure 7a. CBF h is nonnegative throughout the maneuver, and λ is positive at times when the desired control forces and torques are modified.

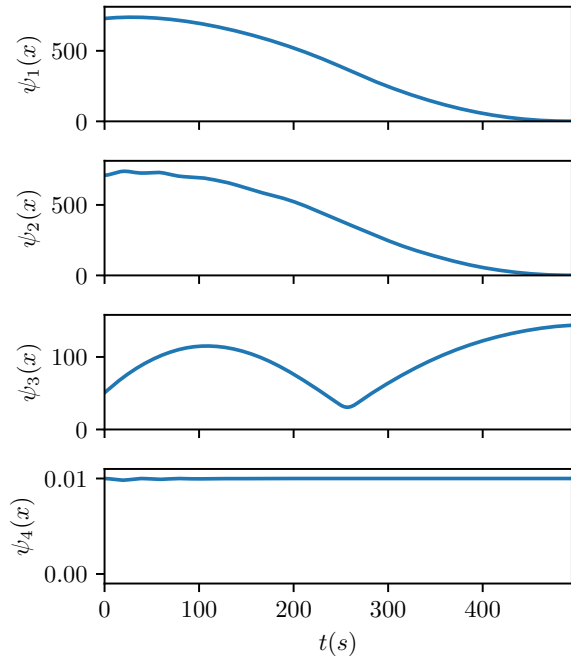


Figure 7b. Barrier functions are nonnegative.

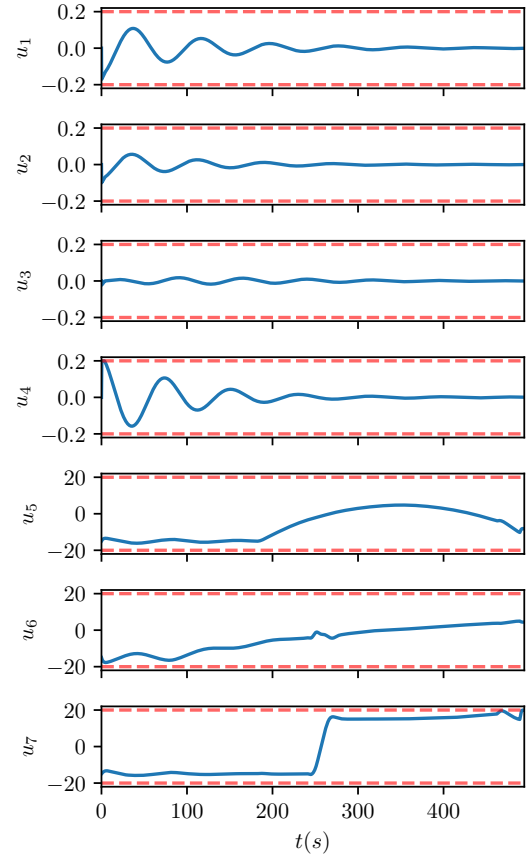


Figure 7c. Actuator inputs are in the admissible set (dashed lines).

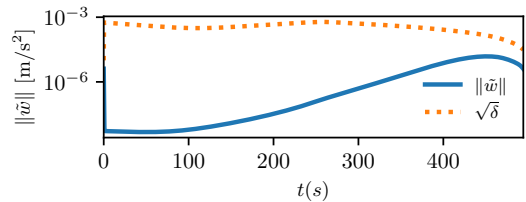


Figure 7d. Estimate error $\|\tilde{w}\|$ is less than the computable upper bound $\sqrt{\delta}$.

Figure 7. CBF behavior for the robotic lander and an ellipsoidal asteroid.

EXAMPLE 3. We again consider the robotic lander from Example 2, but the asteroid is irregularly shaped with nonuniform density, which is modeled as the union of 2 uniform-density ellipsoids as shown in Figure 8a. One ellipsoid has semi-axes lengths of 1000 m, 400 m, and 400 m, and density $\rho_1 = 1460 \text{ kg/m}^3$. The other has semi-axes lengths of 800 m, 400 m, and 400 m, and density $\rho_2 = 1380 \text{ kg/m}^3$. The asteroid rotates about its principal axis associated with the largest moment of inertia, with synodic period 4 h.

The initial conditions are $r_0 = [1554.259 \ -550.0 \ -710.605]^\top \text{ m}$, $v_0 = [-1 \ 4 \ -2]^\top \text{ m/s}$, $\omega_0 = 0 \text{ rad/s}$, and R_0 is the rotation corresponding to a 1-2-3 Euler sequence with angles (0.174, 0.087, 0) rad. The final position is $r_f = [762.523 \ 0 \ 430.952]^\top \text{ m}$, and the final attitude R_f is the rotation corresponding to a 1-2-3 Euler sequence with angles $(\pi, 1.13, -\pi/2) \text{ rad}$.

Landing is achieved at $t_l = 495.4 \text{ s}$ with $\|r - r_f\| = 1.582 \text{ m}$, $\|v - v_f\| = 0.148 \text{ m/s}$, $d(R, R_f) = 0.001 \text{ rad}$, and $\|\omega - \omega_f\| = 0.002 \text{ rad/s}$. Figure 8a shows the spacecraft trajectory, and Figure 8b shows $r - r_f$ and $d(R, R_f)$. Figure 9a shows that h is nonnegative throughout the maneuver, which implies that $\psi_1, \psi_2, \psi_3, \psi_4, \phi_1, \dots, \phi_{18}$ are nonnegative. Thus, $x(t) \in \Psi$ and $u(t) \in \mathcal{U}$ for $t \in [0, t_l]$. Figure 9a also shows that λ is positive for $t \in [0, 98]$ and $t \in [456, 495.4]$. In comparison to Figure 7a, there are 2 periods of modification to the desired control forces and torques because of the gravitational effects not captured by the known model G_m . Figures 9b and 9c confirm that the state and actuator constraints are satisfied, and Figure 9d shows that the magnitude of the estimation error $\tilde{w}$ and satisfies the upper bound $\sqrt{\delta}$.

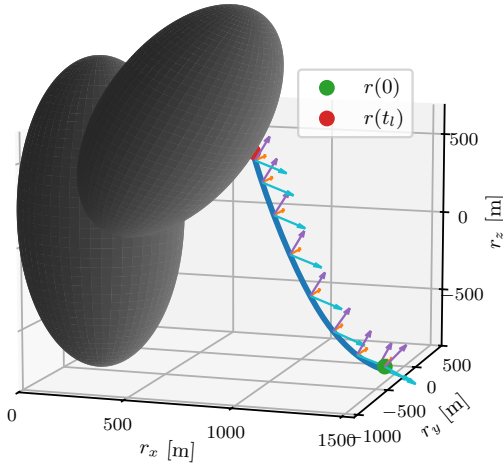


Figure 8a. Spacecraft trajectory and body-frame attitude.

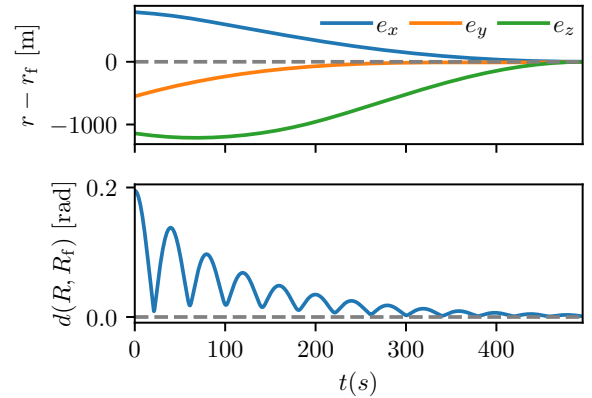


Figure 8b. Distance $r - r_f$ to landing, and geodesic distance $d(R, R_f)$ to landing attitude. Note e_x, e_y, e_z are the entries of $r - r_f$.

Figure 8. Robotic lander soft-landing maneuver on irregularly-shaped asteroid.

EXAMPLE 4. We consider a crewed landing module similar to the Apollo Landing Module (Grumman Aerospace Corporation, 1969) with mass of $m = 3500 \text{ kg}$ and inertia matrix

$$J = \begin{bmatrix} 4200 & 80 & -40 \\ 80 & 4000 & -60 \\ -40 & -60 & 3200 \end{bmatrix} \text{ kg} \cdot \text{m}^2.$$

The spacecraft has 8 bi-directional thrusters, which are arranged according to (Grumman Aerospace Corporation, 1969, Section 3-5) and has the allocation matrix

$$M = \begin{bmatrix} 1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

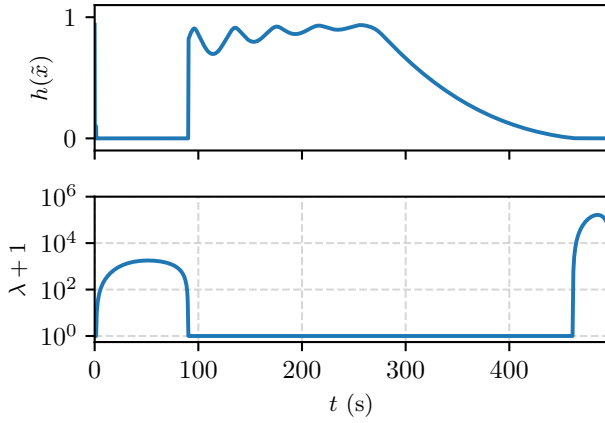


Figure 9a. CBF h is nonnegative throughout the maneuver, and λ is positive at times when the desired control is modified.

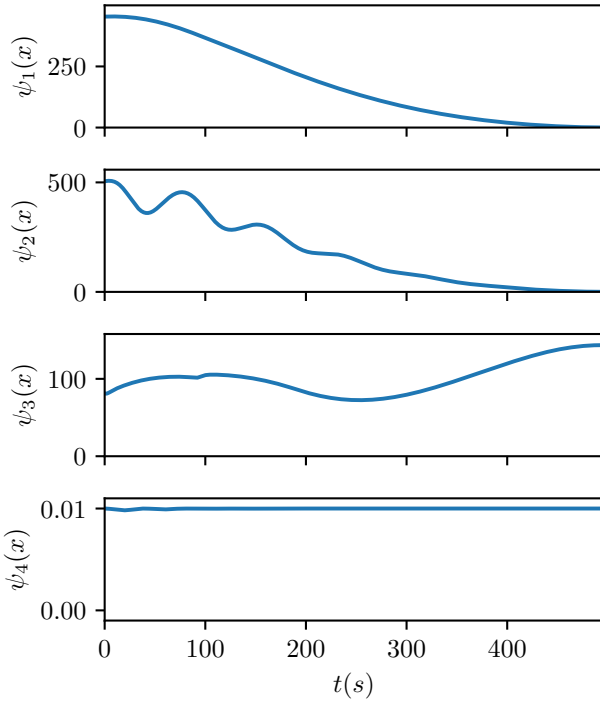


Figure 9b. Barrier functions are nonnegative.

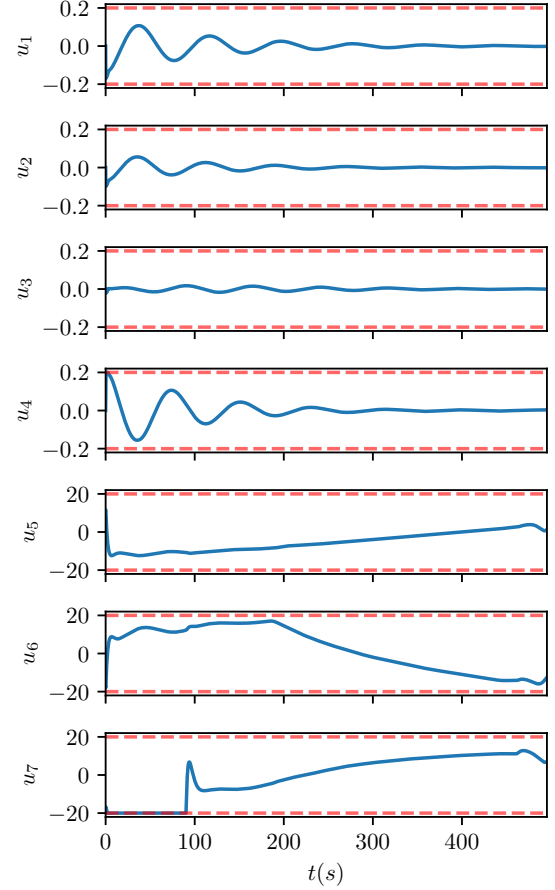


Figure 9c. Actuator inputs are in the admissible set (dashed lines).

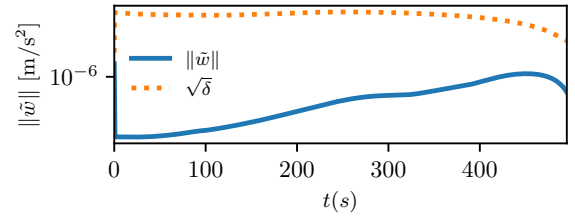


Figure 9d. Estimate error $\|\tilde{w}\|$ is less than the computable upper bound $\sqrt{\delta}$.

Figure 9. CBF behavior for the robotic lander and an irregularly-shaped asteroid.

Each thruster has a force limit of ± 100 N. Thus, the admissible actuator input set $\mathcal{U}$ is a hyper-cube with 16 half-spaces, and the resulting admissible control-force-and-torque set Φ has 24 half-spaces. Control gains are $K_r = 0.1 I_3$, $K_v = 0.632 I_3$, $K_R = \text{diag}(189, 180, 144) \cdot 10^4$, and $K_\omega = \text{diag}(63, 59.4, 46.8) \cdot 10^4$. We use the asteroid, initial conditions, and final conditions from Example 3.

Landing is achieved at $t_1 = 469.4$ s, where $\|r - r_f\| = 0.786$ m, $\|v - v_f\| = 0.147$ m/s, $d(R, R_f) = 0.001$ rad, $\|\omega - \omega_f\| = 0.001$ rad/s. Figure 10a shows the spacecraft trajectory, and Figure 10b shows $r - r_f$ and $d(R, R_f)$. Figure 11a shows that h is nonnegative throughout the maneuver, which implies that $\psi_1, \psi_2, \psi_3, \psi_4, \phi_1, \dots, \phi_{24}$ are nonnegative. Thus, $x(t) \in \Psi$ and $u(t) \in \mathcal{U}$ for $t \in [0, t_1]$. Figure 11a also shows that λ is positive for $t \in [450, 469.4]$. Figures 11b and 11c confirm that the state and actuator

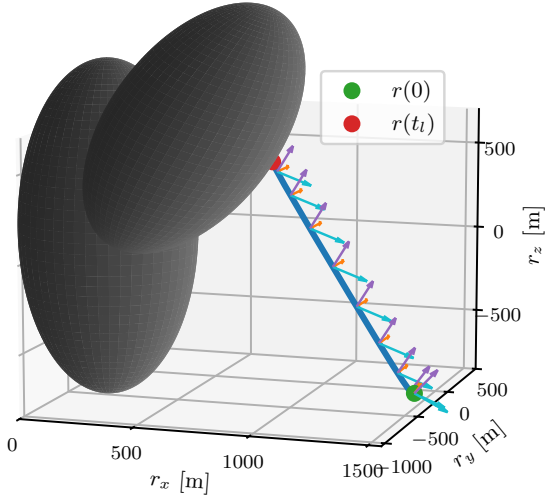


Figure 10a. Spacecraft trajectory and body-frame attitude.

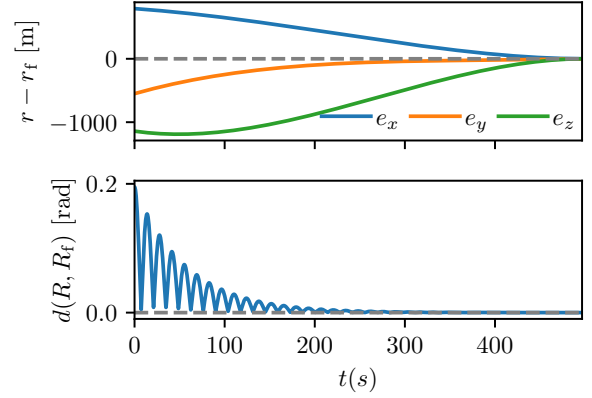


Figure 10b. Distance $r - r_f$ to landing, and geodesic distance $d(R, R_f)$ to landing attitude. Note e_x, e_y, e_z are the entries of $r - r_f$.

Figure 10. Crewed landing module soft-landing maneuver on irregularly-shaped asteroid.

constraints are satisfied throughout the maneuver. The gravity estimator has similar performance to Figure 9d.

8 CONCLUSIONS

This article presents a control architecture for safe soft-landing maneuvers of overactuated and fully actuated spacecraft on small celestial bodies under gravitational uncertainty. Several key contributions enable simultaneous position and attitude control while satisfying state and input constraints.

The constraint projection method presented in Section 4 is a key enabler to re-frame the actuator constraint $u \in \mathcal{U}$ as a constraint on the forces and torques $\mu \in \Phi$. Thus, allowing to focus on designing control forces and torques instead of individual actuator commands. Moreover, the allocation presented in (18) provides a mapping from the forces and torques to the actuator inputs given by a quadratic program which Proposition 1 proves to be feasible and have a unique solution. It is worth noting that other allocation algorithms may be used to perform such mapping provided that they enforce the constraint $u \in \mathcal{U}$.

The gravitational uncertainty estimation $\hat{w}$ presented in Section 5 is an essential element of the control architecture. It is not only used to compensate for the disturbance in (24). But also, by leveraging on δ and Proposition 2, we design a computable lower bound of the CBF constraint function (44) as shown in Proposition 7. Finally in Section 6, we present a minimum-intervention control which is shown to be continuous in Proposition 9 and Theorem 1 shows that it satisfies (O2) and (O3) while producing a control that approximates the desired control (26), which is designed to achieve (O1).

The simulation results demonstrate that the proposed method enables successful landings under significant gravitational uncertainty where traditional feedback control fails. Furthermore, the closed-form nature of the controller ensures the computational efficiency required for autonomous onboard implementation.

A limitation of the polytope projection method is the growth in the number of half-spaces ℓ , which depends on the geometry of $\mathcal{U}$ and the allocation matrix M . We note that the projection of any convex set through the affine mapping (9) is a convex set, indicating that the input constraint projection method can be extended to more general convex sets. However, finding efficient representations for Φ when $\mathcal{U}$ is a non-polytopic convex set is open for future work.

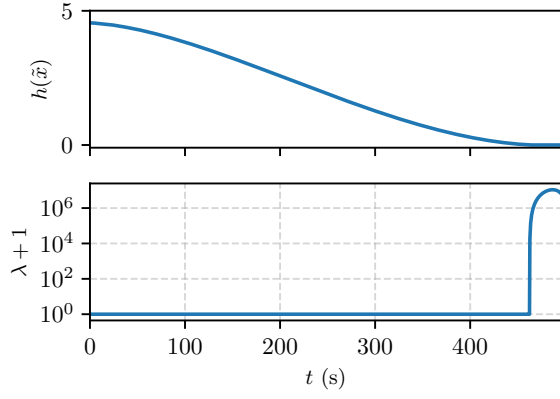


Figure 11a. CBF h is nonnegative throughout the maneuver, and λ is positive at times when the desired control is modified.

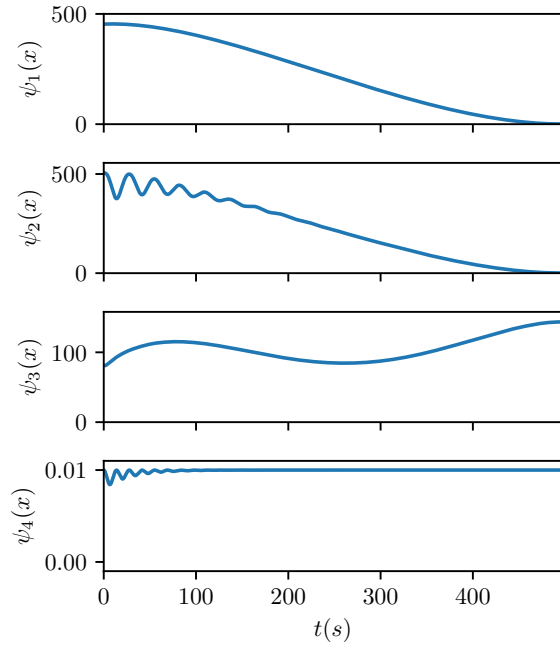


Figure 11b. State barrier functions are nonnegative.

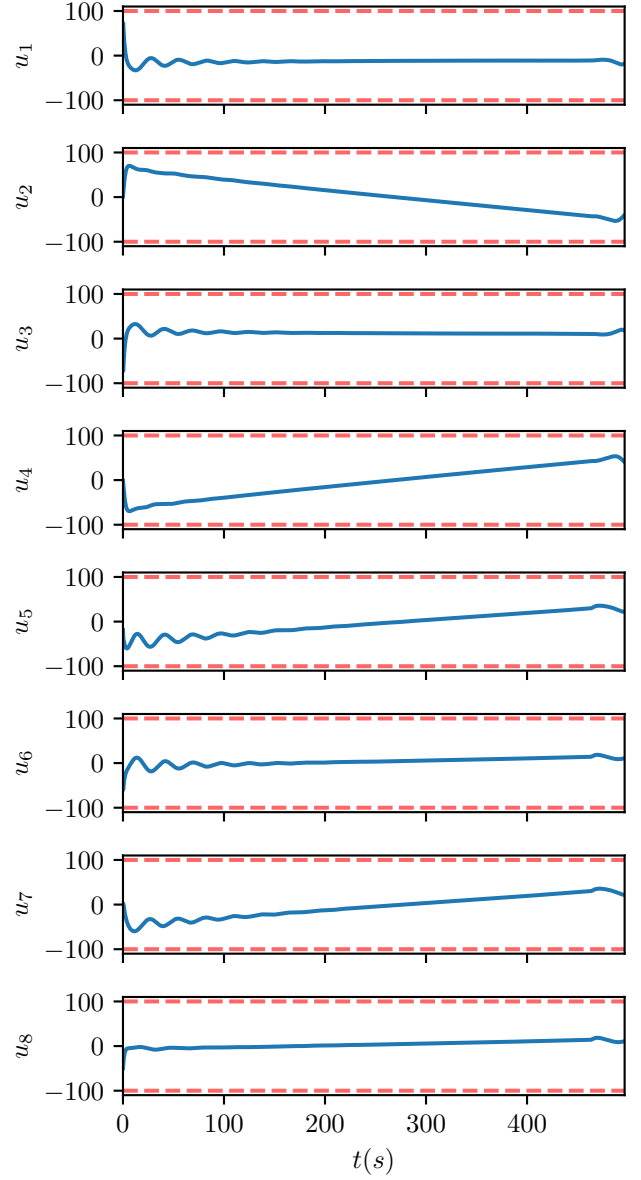


Figure 11c. Actuator inputs are in the admissible set (dashed lines).

Figure 11. CBF behavior for the crewed landing module and an irregularly-shaped asteroid.

APPENDIX A PROOFS OF PROPOSITION 2 AND 3

PROOF OF PROPOSITION 2. Since $\tilde{w}(x, \nu) = w(x) - \tau v - \nu$, it follows that the time derivative of (23) along (5) to (8), (19) and (22) is

$$\begin{aligned}\dot{\xi} &= \dot{\delta} - 2\tilde{w}^\top(\dot{w} - \tau\dot{v} - \dot{\nu}) \\ &= -\tau\delta + \frac{c_\delta}{\tau}\|v\|^2 - 2\tilde{w}^\top\dot{w} + 2\tau\|\tilde{w}\|^2.\end{aligned}\tag{51}$$

Since $0 \leq \|\sqrt{\tau}\tilde{w} - \dot{w}/\sqrt{\tau}\|^2 = \tau\|\tilde{w}\|^2 - 2\tilde{w}^\top \dot{w} + \|\dot{w}\|^2/\tau$, it follows that

$$\tau\|\tilde{w}\|^2 - 2\tilde{w}^\top \dot{w} \geq -\frac{\|\dot{w}\|^2}{\tau},$$

which combined with (51) implies that

$$\dot{\xi} \geq -\tau\delta + \frac{c_\delta}{\tau}\|v\|^2 - \frac{\|\dot{w}\|^2}{\tau} + \tau\|\tilde{w}\|^2. \quad (52)$$

Next, since $w(x) = \Delta(r)$, it follows that $\dot{w} = \Delta'(r)v$. Moreover, it follows from Assumption 1 that there exists $c_\delta > 0$ such that $c_\delta \geq \|\Delta'(r)\|^2$. Thus, it follows that $\|\dot{w}\|^2 \leq \|\Delta'(r)\|^2\|v\|^2 \leq c_\delta\|v\|^2$, which combined with (52) yields

$$\dot{\xi} \geq -\tau\delta + \tau\|\tilde{w}\|^2 = -\tau\xi. \quad (53)$$

Since $\xi(x_0, \nu_0, \delta_0) \geq 0$, it follows from (53) that for all $t \geq 0$, $\xi(x(t), \nu(t), \delta(t)) \geq 0$. ■

PROOF OF PROPOSITION 3. It follows from (5) to (8), (19) and (22) that

$$\dot{z}(t) = Az(t) + B\tilde{w}(x(t), \nu(t)), \quad (54)$$

where

$$z(t) \triangleq \begin{bmatrix} e_r(t, x(t)) \\ e_v(t, x(t)) \end{bmatrix}, \quad A = \begin{bmatrix} 0_{3 \times 3} & I_{3 \times 3} \\ -K_r & -K_v \end{bmatrix}, \quad B = \begin{bmatrix} 0_{3 \times 3} \\ I_{3 \times 3} \end{bmatrix}.$$

Since K_r and K_v are positive definite, it follows that A is asymptotically stable. Let $P \in \mathbb{R}^{6 \times 6}$ be the positive-definite solution to $A^\top P + PA = -I$, and define $V(t) \triangleq z(t)^\top Pz(t)$. The time derivative of V along the trajectories of (54) is

$$\dot{V} = -z^\top z + 2z^\top PB\tilde{w}. \quad (55)$$

Next, let $\varsigma \in (0, 1)$ be such that $\varsigma^2 \neq 1 - \tau\lambda_{\max}(P)$, and it follows from (55) that

$$\begin{aligned} \dot{V} &= -z^\top z - \left(\varsigma z - \frac{1}{\varsigma}PB\tilde{w}\right)^\top \left(\varsigma z - \frac{1}{\varsigma}PB\tilde{w}\right) + \varsigma^2 z^\top z + \frac{1}{\varsigma^2}\tilde{w}^\top B^\top PPB\tilde{w}, \\ &\leq -(1 - \varsigma^2)z^\top z + \frac{\lambda_{\max}(B^\top PPB)}{\varsigma^2}\tilde{w}^\top \tilde{w}. \end{aligned} \quad (56)$$

Since $V \leq \lambda_{\max}(P)z^\top z$, it follows that $-z^\top z \leq -V/\lambda_{\max}(P)$, which combined with (56) yields

$$\dot{V} \leq -\frac{(1 - \varsigma^2)}{\lambda_{\max}(P)}V + \frac{\lambda_{\max}(B^\top PPB)}{\varsigma^2}\tilde{w}^\top \tilde{w}. \quad (57)$$

Since $\xi(x_0, \nu_0, \delta_0) \geq 0$, Proposition 2 and (57) imply that

$$\dot{V} \leq -\frac{(1 - \varsigma^2)}{\lambda_{\max}(P)}V + \frac{\lambda_{\max}(B^\top PPB)}{\varsigma^2}\delta. \quad (58)$$

Since the solution to (22) is

$$\begin{aligned}\delta(t) &= \delta_0 e^{-\tau t} + \frac{c_\delta}{\tau} \int_0^t e^{-\tau(t-s)} \|v(s)\|^2 ds \\ &\leq \delta_0 e^{-\tau t} + \frac{c_\delta}{\tau} \int_0^t e^{-\tau(t-s)} v_m^2 ds, \\ &= \delta_0 e^{-\tau t} + \frac{c_\delta v_m^2}{\tau^2} (1 - e^{-\tau t}),\end{aligned}$$

substituting into (58) yields

$$\dot{V}(t) \leq -c_1 V(t) + \frac{c_2}{\tau^2} + c_3 e^{-\tau t}, \quad (59)$$

where

$$c_1 = \frac{(1 - \varsigma^2)}{\lambda_{\max}(P)}, \quad c_2 = \frac{\lambda_{\max}(B^\top P P B) c_\delta v_m^2}{\varsigma^2}, \quad c_3 = \left(\delta_0 - \frac{c_\delta v_m^2}{\tau^2}\right) \frac{\lambda_{\max}(B^\top P P B)}{\varsigma^2}.$$

Next, consider $\bar{V} : [0, \infty) \rightarrow \mathbb{R}$ that satisfies

$$\dot{\bar{V}}(t) = -c_1 \bar{V}(t) + \frac{c_2}{\tau^2} + c_3 e^{-\tau t}, \quad (60)$$

where $\bar{V}(0) = V(0)$. Since $\varsigma^2 \neq 1 - \tau \lambda_{\max}(P)$, it follows that $c_1 \neq \tau$, which implies that the solution to (60) is

$$\bar{V}(t) = \bar{V}(0) e^{-c_1 t} + \frac{c_2}{c_1 \tau^2} (1 - e^{-c_1 t}) + \frac{c_3}{c_1 - \tau} (e^{-\tau t} - e^{-c_1 t}). \quad (61)$$

Using the comparison lemma (Khalil, 2002, Lemma 3.4) on (59) and (60) yields $V(t) \leq \bar{V}(t)$. Since, in addition, $\lambda_{\min}(P) z^\top z \leq V$, it follows that for all $t \geq 0$, $z(t)^\top z(t) \leq \bar{V}(t) / \lambda_{\min}(P)$. Finally, since (61) implies that $\lim_{t \rightarrow \infty} \bar{V}(t) = \frac{c_2}{c_1 \tau^2}$, letting $\vartheta \triangleq \sqrt{\frac{c_2}{c_1 \lambda_{\min}(P)}}$ yields that $\|z(t)\| \rightarrow [0, \frac{\vartheta}{\tau}]$ as $t \rightarrow \infty$. ■

REFERENCES

- Alan, A., Molnar, T. G., Ames, A. D., and Orosz, G. (2025). Generalizing Robust Control Barrier Functions From a Controller Design Perspective. *IEEE Open Journal of Control Systems* 4, 54–69. doi:10.1109/OJCSYS.2025.3529364
- Alan, A., Molnar, T. G., Das, E., Ames, A. D., and Orosz, G. (2023). Disturbance Observers for Robust Safety-Critical Control With Control Barrier Functions. *IEEE Control Systems Letters* 7, 1123–1128. doi:10.1109/LCSYS.2022.3232059
- Ames, A. D., Coogan, S., Egerstedt, M., Notomista, G., Sreenath, K., and Tabuada, P. (2019). Control Barrier Functions: Theory and Applications. In *2019 18th European Control Conference (ECC)* (Naples, Italy: IEEE), 3420–3431. doi:10.23919/ECC.2019.8796030
- Arenas-Urbe, F. (2026). Higher-Order Gravitational Models: A Tutorial on Spherical Harmonics and the Newtonian Model. doi:10.48550/arXiv.2601.17628. ArXiv:2601.17628
- Arenas-Urbe, F., Seigler, T. M., and Hoagg, J. B. (2025). Safe Landing on Small Celestial Bodies with Gravitational Uncertainty Using Disturbance Estimation and Control Barrier Functions. doi:10.48550/ARXIV.2510.05895
- Breeden, J. and Panagou, D. (2023). Robust Control Barrier Functions under high relative degree and input constraints for satellite trajectories. *Automatica* 155, 111109. doi:10.1016/j.automatica.2023.111109

- Bull, R., Mitch, R., Atchison, J., McMahon, J., Rivkin, A., and Mazarico, E. (2021). Optical Gravimetry mass measurement performance for small body flyby missions. *Planetary and Space Science* 205, 105289. doi:10.1016/j.pss.2021.105289
- Das, E. and Murray, R. M. (2022). Robust Safe Control Synthesis with Disturbance Observer-Based Control Barrier Functions. In *2022 IEEE 61st Conference on Decision and Control (CDC)* (Cancun, Mexico: IEEE), 5566–5573. doi:10.1109/CDC51059.2022.9993032
- Daş, E. and Burdick, J. W. (2025). Robust Control Barrier Functions Using Uncertainty Estimation With Application to Mobile Robots. *IEEE Transactions on Automatic Control* 70, 4766–4773. doi:10.1109/TAC.2025.3538742
- Dunham, W., Petersen, C., and Kolmanovsky, I. (2016). Constrained control for soft landing on an asteroid with gravity model uncertainty. In *2016 American Control Conference (ACC)* (Boston, MA, USA: IEEE), 5842–5847. doi:10.1109/ACC.2016.7526585
- Durham, W. C. (1994). Attainable moments for the constrained control allocation problem. *Journal of Guidance, Control, and Dynamics* 17, 1371–1373. doi:10.2514/3.21360
- Elango, P., Vinod, A., Di Cairano, S., and Weiss, A. (2025). Continuous-Time Successive Convexification for Passively-Safe Six-Degree-of-Freedom Powered-Descent Guidance. In *AIAA SCITECH 2025 Forum* (Orlando, FL: American Institute of Aeronautics and Astronautics), 1. doi:10.2514/6.2025-1894
- Fu, J., Chen, L., Zhang, D., and Shao, X. (2022). Orbit–attitude dynamics and control of spacecraft hovering over a captured asteroid in the earth–moon system. *Aerospace Systems* 5, 265–275. doi:10.1007/s42401-021-00122-w
- Fukuda, K. and Prodon, A. (1996). Double description method revisited. In *Combinatorics and Computer Science*, eds. M. Deza, R. Euler, and I. Manoussakis (Berlin, Heidelberg: Springer), 91–111. doi:10.1007/3-540-61576-8_77
- Grumman Aerospace Corporation (1969). *Lunar module - LM10 through LM14: Orientational manual for lunar module vehicle and mission requirements*. Contractor Report NASA-CR-117321; LMA790-2, National Aeronautics and Space Administration (NASA). Document ID: 19710010995; Accession Number: 71N20470; Contract No: NAS9-1100
- Hasan, M. N., Haris, M., and Qin, S. (2022). Fault-tolerant spacecraft attitude control: A critical assessment. *Progress in Aerospace Sciences* 130, 100806. doi:10.1016/j.paerosci.2022.100806
- Ho, T.-M., Jaumann, R., Bibring, J.-P., Grott, M., Glaßmeier, K.-H., Moussi, A., et al. (2021). The MASCOT lander aboard Hayabusa2: The in-situ exploration of NEA (162173) Ryugu. *Planetary and Space Science* 200, 105200. doi:10.1016/j.pss.2021.105200
- Jankovic, M. (2018). Robust control barrier functions for constrained stabilization of nonlinear systems. *Automatica* 96, 359–367. doi:10.1016/j.automatica.2018.07.004
- Kamat, S. S., Seigler, T. M., and Hoagg, J. B. (2026). Electromagnetic Formation Flying Using Alternating Magnetic Field Forces and Control Barrier Functions for State and Input Constraints. *IEEE Transactions on Aerospace and Electronic Systems* 62, 6108–6125. doi:10.1109/TAES.2026.3664815
- Khalil, H. K. (2002). *Nonlinear systems* (Upper Saddle River, NJ: Prentice Hall), 3. ed edn.
- Lee, T. (2012). Exponential stability of an attitude tracking control system on SO(3) for large-angle rotational maneuvers. *Systems & Control Letters* 61, 231–237. doi:10.1016/j.sysconle.2011.10.017
- Malyuta, D., Yu, Y., Elango, P., and Açıkmeşe, B. (2021). Advances in trajectory optimization for space vehicle control. *Annual Reviews in Control* 52, 282–315. doi:10.1016/j.arcontrol.2021.04.013
- Mammarella, M., D’Amico, S., Pavone, M., Linares, R., Acheson, M. J., Ankersen, F., et al. (2025). Controls for Space: a perspective to 2030s and beyond I. In *2025 American Control Conference (ACC)* (Denver, CO, USA: IEEE), 2286–2296. doi:10.23919/ACC63710.2025.11107678

- Modell, S. (2025). Space mining as an emerging organisational field: An institutional logics perspective. *Space Policy* 74, 101712. doi:10.1016/j.spacepol.2025.101712
- Ong, P. and Ames, A. D. (2023). Intermittent Safety Filters for Event-Triggered Safety Maneuvers with Application to Satellite Orbit Transfers. In *2023 62nd IEEE Conference on Decision and Control (CDC)* (Singapore, Singapore: IEEE), 870–877. doi:10.1109/CDC49753.2023.10383437
- Peñarroya, P. and Paoli, R. (2019). Orbit propagation around small bodies using spherical harmonic coefficients obtained from polyhedron shape models. *Proceedings of the International Astronomical Union* 15, 203–210. doi:10.1017/S1743921321001496
- Rabiee, P. and Hoagg, J. B. (2024a). A Closed-Form Control for Safety Under Input Constraints Using a Composition of Control Barrier Functions. doi:10.48550/ARXIV.2406.16874
- Rabiee, P. and Hoagg, J. B. (2024b). Composition of Control Barrier Functions with Differing Relative Degrees for Safety Under Input Constraints. In *2024 American Control Conference (ACC)* (Toronto, ON, Canada: IEEE), 3692–3697. doi:10.23919/ACC60939.2024.10644151
- Rabiee, P. and Hoagg, J. B. (2025). Soft-minimum and soft-maximum barrier functions for safety with actuation constraints. *Automatica* 171, 111921. doi:https://doi.org/10.1016/j.automatica.2024.111921
- Reynolds, T. P., Szmuk, M., Malyuta, D., Mesbahi, M., Açıkmeşe, B., and Carson, J. M. (2020). Dual Quaternion-Based Powered Descent Guidance with State-Triggered Constraints. *Journal of Guidance, Control, and Dynamics* 43, 1584–1599. doi:10.2514/1.G004536
- Rockafellar, R. T. (2015). *Convex Analysis*. Princeton Landmarks in Mathematics and Physics (Princeton: Princeton University Press)
- Safari, A. and Hoagg, J. B. (2025). Time-Varying Soft-Maximum Barrier Functions for Safety in Unmapped and Dynamic Environments. *IEEE Transactions on Control Systems Technology* 33, 2255–2270. doi:10.1109/TCST.2025.3587198
- Sagliano, M., Seelbinder, D., Theil, S., and Lu, P. (2024). Six-Degree-of-Freedom Rocket Landing Optimization via Augmented Convex–Concave Decomposition. *Journal of Guidance, Control, and Dynamics* 47, 20–35. doi:10.2514/1.G007570
- Scheeres, D. J. (2012). *Orbital Motion in Strongly Perturbed Environments: Applications to Asteroid, Comet and Planetary Satellite Orbiters* (Berlin, Heidelberg: Springer Berlin Heidelberg). doi:10.1007/978-3-642-03256-1
- Simplício, P., Marcos, A., Joffre, E., Zamaro, M., and Silva, N. (2018). Synthesis and analysis of robust control compensators for Space descent & landing. *International Journal of Robust and Nonlinear Control* 28, 3871–3892. doi:10.1002/rnc.4109
- Sun, C., Xiao, Y., Sun, Z., and Ye, D. (2021). Dual Quaternion Based Close Proximity Operation for In-Orbit Assembly via Model Predictive Control. *International Journal of Aerospace Engineering* 2021, 1–14. doi:10.1155/2021/1305095
- Sun, J., Yang, J., and Zeng, Z. (2024). Safety-Critical Control With Control Barrier Function Based on Disturbance Observer. *IEEE Transactions on Automatic Control* 69, 4750–4756. doi:10.1109/TAC.2024.3352707
- Sun, L. and Zheng, Z. (2018). Adaptive relative pose control of spacecraft with model couplings and uncertainties. *Acta Astronautica* 143, 29–36. doi:10.1016/j.actaastro.2017.11.006
- Tan, X., Shaw Cortez, W., and Dimarogonas, D. V. (2022). High-Order Barrier Functions: Robustness, Safety, and Performance-Critical Control. *IEEE Transactions on Automatic Control* 67, 3021–3028. doi:10.1109/TAC.2021.3089639
- Tiwari, M., Coyle, E., and Prazenica, R. J. (2022). Direct-Adaptive Nonlinear MPC for Spacecraft Near Asteroids. *Aerospace* 9, 159. doi:10.3390/aerospace9030159

- Tiwari, M., Prazenica, R., and Henderson, T. (2023). Direct adaptive control of spacecraft near asteroids. *Acta Astronautica* 202, 197–213. doi:10.1016/j.actaastro.2022.10.014
- Van Leeuwen, S., Skibik, T., Nicotra, M., Kolmanovsky, I., and Liao-McPherson, D. (2022). A Nonlinear Predictive Control Strategy for Landing on an Asteroid. In *2022 American Control Conference (ACC)* (Atlanta, GA, USA: IEEE), 443–449. doi:10.23919/ACC53348.2022.9867552
- Wang, X., Yang, J., Liu, C., Yan, Y., and Li, S. (2025). Safety-Critical Disturbance Rejection Control of Nonlinear Systems With Unmatched Disturbances. *IEEE Transactions on Automatic Control* 70, 2722–2729. doi:10.1109/TAC.2024.3496572
- Wang, Y. and Ji, H. (2019). Integrated relative position and attitude control for spacecraft rendezvous with ISS and finite-time convergence. *Aerospace Science and Technology* 85, 234–245. doi:10.1016/j.ast.2018.12.005
- Wieland, P. and Allgöwer, F. (2007). Constructive safety using control barrier functions. *IFAC Proceedings Volumes* 40, 462–467. doi:https://doi.org/10.3182/20070822-3-ZA-2920.00076. 7th IFAC Symposium on Nonlinear Control Systems
- Zhang, F. and Duan, G. (2023). Coupled Dynamics and Integrated Control for Position and Attitude Motions of Spacecraft: A Survey. *IEEE/CAA Journal of Automatica Sinica* 10, 2187–2208. doi:10.1109/JAS.2023.123306
- Zhang, H., Ye, D., Xiao, Y., and Sun, Z. (2022). Adaptive Control on SE(3) for Spacecraft Pose Tracking With Harmonic Disturbance and Input Saturation. *IEEE Transactions on Aerospace and Electronic Systems* 58, 4578–4594. doi:10.1109/TAES.2022.3165524
- Ziegler, G. M. (1995). *Lectures on polytopes*. No. 152 in Graduate texts in mathematics (Berlin New York: Springer)